

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

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Chemistry
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AIOB

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Table of Contents

Ch	Topic	Pages
1	Atomic Structure	03-26
2	Wave Mechanical Concept of Atom	27-55
3	Modern Periodic Table	56-76
4	Chemical Reactions	77-96
5	Chemical Bonds	97-112
6	Molecular Orbital Theory (MOT)	113-125

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Atomic Structure
Date: _____

ATOMIC
Structure
Chapter

Atomic Structure:—

Atomic theory of Dalton:—

Q:- State Dalton Atomic Theory:-

⇒ ① All matter is composed of tiny particles called atom which is not created, destroyed or split.

(-ମାତ୍ର - ପଦାର୍ଥ ପରମାଣୁ ନାମକ ଓଡ଼ିଆ କଣ ଦ୍ଵାରା ଗଠିତ ଥାଏ, ସୃଷ୍ଟି, ନଷ୍ଟ ହେବା ବା ବିଭକ୍ତ ହେବା ପାଏ ନା,)

② Atoms of the same element are identical in mass and chemical properties.

(-ଏକତ୍ର - ଗୁଣିତ - ପରମାଣୁ - ଓଡ଼ିଆ ଶାରୀରିକ - ରାସାୟନିକ ଗୁଣ ସମାନ)

③ Compounds are formed by the combination of atoms from different

(-ବିଭିନ୍ନ ଉପାଦାନ ଓଡ଼ିଆ - ପରମାଣୁ ମିଶ୍ରିତ ହୋଇ ଗଠିତ ହୁଏ)

④ Chemical reactions involve the rearrangement of atoms.

(ରାସାୟନିକ ବିକିରଣ - ପରମାଣୁ - ପୁନର୍ଗଠନ - ଓଡ଼ିଆ)

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Q Contribution:-

⇒ Dalton provided the basic theory that all matter - whether element, compound, or mixture is composed of small particles called atoms.

Q Limitations:-

⇒ ① Atoms can be divided into subatomic particles, namely electrons, protons, & neutrons.

② Not all atoms of the same element are identical. they can have different masses and chemical properties. this is known as isotopes.

Q Rutherford's Model

Q Explain Rutherford's atomic model also explain if there any Limitations:-?

⇒ Atomic Model:-

① At the center of a atom have a small dense core, called the nucleus.

(-प्रमाणित - कोश - एनर्जी - कोर, एन - कोर - एनर्जी - कोर - निरूपित - एन २०,) which containing all of the atom's mass. (-एन प्रमाणित एन २० २००)

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

and rest of the atom mostly empty (ସବୁ ଖାଲି ସ୍ଥାନ)
(ବିଶାଳତା ସାମି-ସାମି)

② The nucleus holds the all positive charge of the atom, while electrons are dispersed in empty space surrounding it.

③ The electrons were moving in orbits around the nucleus, like planets around the sun.

(e^- ଇଲେକ୍ଟ୍ରନ୍ ସବୁ ନ୍ୟୁକ୍ଲିୟସ୍ ଚାରିପାଖରେ ଘୁରୁଥିଲା ଯେପରିକି ଖଗୋଳ ଗ୍ରହ ଯେପରିକି ସୂର୍ଯ୍ୟ ଚାରିପାଖରେ ଘୁରୁଥାଏ)।

Limitations or Weakness:-

⇒ ① This atomic model failed to explain the stability of atoms.

② This model of the atom failed to explain the existence of definite lines in the hydrogen spectrum.

③ Imagine electrons going around the nucleus like a tiny planet. But they can't keep doing that forever because they lose energy and eventually crash into the nucleus.

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Contribution:-

⇒ Rutherford laid the foundation of the model picture of atom.

Subatomic particles:-

⇒ There are three subatomic particles

	<u>Mass</u>	<u>Units</u>	<u>Coulombs</u>
① Electron, e^-	$9.109 \times 10^{-31} \text{ kg}$	-1	$-1.602 \times 10^{-19} \text{ C}$
② Proton, p^+	$1.67 \times 10^{-27} \text{ kg}$	+1	$+1.602 \times 10^{-19} \text{ C}$
③ Neutron, n/n^0	$1.67 \times 10^{-27} \text{ kg}$	0	0

Atomic Number: (परमाणु क्रमांक)

⇒ The atomic number of protons in the nucleus of an atom.

mass Number (द्रव्य क्रमांक) :-

⇒ The mass Number is total Number of protons and neutrons in a nucleus.

Isotopes :-

⇒ Isotopes are atoms of the same element that have the same number of protons but a different number of neutrons,

Electromagnetic Radiation

⇒ It is a form of energy that propagates as both both electrical and magnetic waves travelling in packet of energy called photons.

Some forms of electromagnetic radiations are,

- ① Radio waves,
- ② Visible light,
- ③ Infrared light,
- ④ UV light
- ⑤ X-rays. etc.

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

The uses of EMP:-

- ① X-ray → are used in medicine.
- ② UV-ray → lead to sunburns.
- ③ Radio wave → For communication.
- ④ Visible light → for optic communication, photography.

Characteristics of waves:-

Wavelength: (λ):-

⇒ Distance between identical points of two consecutive wave. represent by ' λ '.

Unit:- cm, m, Å

$$1 \text{ Å} = 10^{-10} \text{ m}$$

Frequency (ν, n):-

⇒ The Number of complete cycle per sec.
represent by ' ν/n '

Unit:- $\text{Hz} / \text{sec}^{-1}$

Speed (c) :-

⇒ The speed of a wave is the distance through which a particular wave travels in one sec.

Represent by 'c'

Unit :- m/s

$$c = \lambda \times \nu$$

Wave Number ($\bar{\nu}$ / nu bar) :-

$$\bar{\nu} = \frac{1}{\lambda}$$

Note :-

⇒ A wave of high frequency has a shorter wavelength, while a wave of low frequency has a longer wavelength.

⇒ An Increase in frequency or a decrease in wavelength represent an increase in energy.

problems:-

⇒ The wavelength of a violet light is 400 nm. Calculate its frequency and wave number. $c = 3 \times 10^8 \text{ m s}^{-1}$

⇒ we know:-

$$c = \nu \lambda$$

$$\therefore \nu = \frac{c}{\lambda}$$

$$= \frac{3 \times 10^8}{400 \times 10^{-9}}$$

$$= 7.5 \times 10^{14} \text{ Hz}$$

Again:-

$$\bar{\nu} = \frac{1}{\lambda}$$

$$= \frac{1}{400 \times 10^{-9}}$$

$$= 2.5 \times 10^6 \text{ m}^{-1}$$

It is given:-

$$\lambda = 400 \text{ nm}$$

$$= 400 \times 10^{-9} \text{ m}$$

$$c = 3 \times 10^8 \text{ m s}^{-1}$$

The Frequency of a strong yellow line in the spectrum of sodium is $5.09 \times 10^{14} \text{ s}^{-1}$. calculate the wavelength of light in nanometers.

⇒ we know:-

$$c = \nu \lambda$$

$$\therefore \lambda = \frac{c}{\nu}$$

$$= \frac{3 \times 10^8}{5.09 \times 10^{14}}$$

$$= 589.39 \text{ nm}$$

It is given:-

$$\nu = 5.09 \times 10^{14} \text{ s}^{-1}$$

$$c = 3 \times 10^8 \text{ m s}^{-1}$$

#Spectrum: (part 1)

⇒ The range of different colours which is produced when light passes through a glass prism.

#Continuous spectrum:-

⇒ A Continuous spectrum contains all wavelengths of light in a certain range.

Note:-

1. The violet component of the spectrum has shorter wavelengths ($4000 - 4250 \text{ \AA}$) and higher frequencies.
2. The red component of the spectrum has longer wavelengths ($6500 - 7500 \text{ \AA}$) and lower frequencies.
3. The Invisible region beyond the violet is called Ultraviolet region. and
4. The one below the red is called Infrared region.

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Atomic Spectra:-

⇒ Atomic spectra refer to the patterns of electromagnetic radiation produced or absorbed by atoms during electron transitions.

Atomic Spectrum of Hydrogen:-

⇒ The atomic spectrum of hydrogen is a key example in understanding atomic structure and quantum mechanics. It consists of distinct lines corresponding to the transitions of electrons between energy levels in a hydrogen atom.

Balmer Series:-

1. A red line where $\lambda = 6563 \text{ \AA}$

2. A blue line where $\lambda = 4861 \text{ \AA}$

3. A blue-green line where $\lambda = 4861 \text{ \AA}$

4. A violet line where $\lambda = 4102 \text{ \AA}$

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Relationship between wavelength and observed line:-

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

λ = wave-length.

R = Rydberg constant = 10967700 m^{-1}
 $= 109677 \text{ cm}^{-1}$

For Balmer series :- $n_1 = 2$

Lyman series :- $n_1 = 1$

Paschen series :- $n_1 = 3$

Brackett series :- $n_1 = 4$

Pfund series :- $n_1 = 5$

For, Lyman series $n_2 = 2, 3, 4, 5, 6, \dots$

Balmer series $n_2 = 3, 4, 5, 6, \dots$

Paschen series $n_2 = 4, 5, 6, 7, \dots$

Brackett series $n_2 = 5, 6, 7, \dots$

Pfund series $n_2 = 6, 7, \dots$

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

5 (Five) Spectral Series:-

⇒ In addition to Balmer series, 4 (Four) other spectral series were discovered in (IR) infrared Ray and (UV) ultra violet regions.

Those are:-

1. Lyman Series (UV)

2. Balmer Series (Visible)

3. Paschen Series (IR)

4. Brackett Series (IR)

5. Pfund Series (IR)

The Energy of a photon:-

$$E = h\nu$$

$$\therefore E = h \times \frac{c}{\lambda}$$

we know,

$$c = \lambda \nu$$

$$\therefore \nu = \frac{c}{\lambda}$$

Where,

h = Planck's Constant = 6.62×10^{-34} J sec.

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Bohr Model of Atom:-

- ⇒ Electrons travel around the Nucleus in specific circular orbits and in no others.
- ⇒ While in these specific orbits, an electron does not lose energy.
- ⇒ An electron can move from one energy level to another by photon jump only.
- ⇒ The Angular momentum (mvr) of an electron orbiting around the nucleus is an integral multiple of Planck's constant divided by 2π .

$$\text{Angular momentum} = mvr = \frac{nh}{2\pi}$$

m = mass of electron.

v = velocity

r = radius.

$n = 1, 2, 3, \dots$

h = Planck's constant.

TOPIC NAME : _____

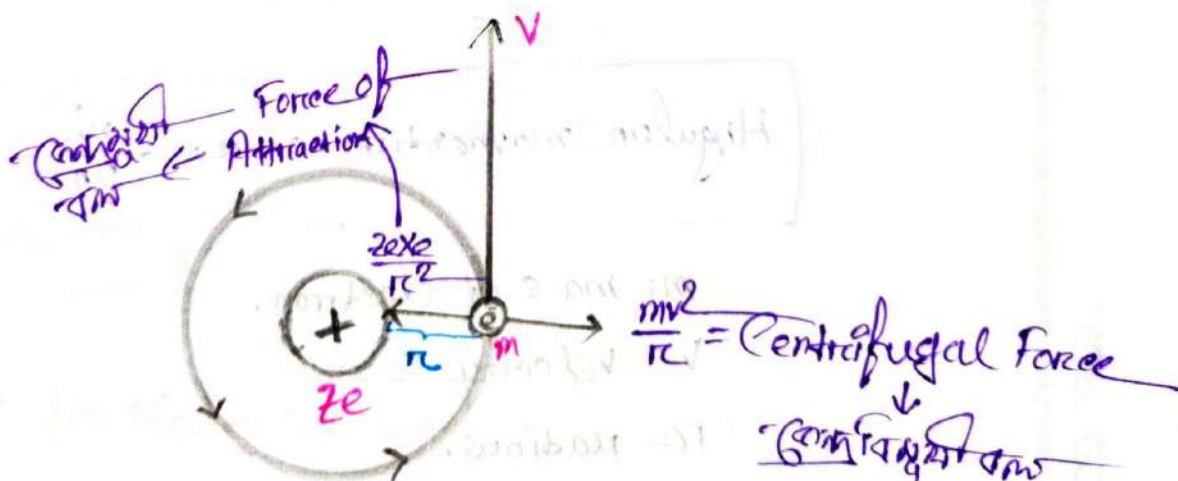
DAY : _____

TIME : _____ DATE : / /

Limitations:

1. Bohr's model doesn't work well for complex atom.
2. It couldn't explain why some spectral lines are more intense than others.
3. It couldn't explain why some spectral lines split into multiple lines in the presence of a magnetic field.

Calculating of Radius of Orbits



→ The force of attraction between the nucleus and the electron (Coulomb's Law)

$$F_1 = \frac{Ze^e}{r^2}$$

where,

Z = Atomic Number

e = Charge on a proton.

r = radius.

→ The centrifugal force acting on the electron.

$$F_2 = \frac{mv^2}{r}$$

Bhor assumed that, force of attraction and centrifugal force of the electron are equal to each other.

$$\therefore F_1 = F_2$$

$$\Rightarrow \frac{Ze^e}{r^2} = \frac{mv^2}{r}$$

$$\Rightarrow \frac{e^2}{r^2} = \frac{mv^2}{r}$$

$$\Rightarrow \frac{e^2}{r} = mv^2 \quad \text{[multiply } r \text{ in both side]} \dots \dots \textcircled{1}$$

According to Bhor's theory, Angular momentum of electrons—

$$mvr = \frac{nh}{2\pi}$$

$$\therefore v = \frac{nh}{2\pi mr} \dots \dots \textcircled{2}$$

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Substituting the value of v in eq (1) $\Rightarrow$

$$\frac{e^2}{r} = m \left(\frac{nh}{2\pi m r} \right)^2$$

$$\Rightarrow \frac{e^2}{r} = \frac{n^2 h^2}{4\pi^2 r^2 m}$$

$$\Rightarrow r = \frac{n^2 h^2}{4\pi^2 e^2 m}$$

$$r_n = n^2 \times 5.3 \times 10^{-11} \text{ m}$$

$n = \text{no. of the orbit}$

$$\begin{cases} h = 6.626 \times 10^{-34} \\ e = 1.6 \times 10^{-19} \\ m = 9.11 \times 10^{-31} \end{cases}$$

#problem :-

$\Rightarrow$ Calculate the first five Bohr radii of the hydrogen atom.

$\Rightarrow$ we know,

$$r_n = n^2 \times 5.3 \times 10^{-11} \text{ m}$$

$\therefore$ 1st orbit radius,

$$\begin{aligned} r_1 &= 1^2 \times 5.3 \times 10^{-11} \text{ m} \\ &= 5.3 \times 10^{-11} \text{ m} \end{aligned}$$

2nd Orbits radius:-

$$r_2 = 2^2 \times 5.3 \times 10^{-11} \text{ m}$$

$$= 2.12 \times 10^{-10} \text{ m}$$

3rd Orbits radius:-

$$r_3 = 3^2 \times 5.3 \times 10^{-11} \text{ m}$$

$$= 4.77 \times 10^{-10} \text{ m}$$

4th Orbits radius:-

$$r_4 = 4^2 \times 5.3 \times 10^{-11} \text{ m}$$

$$= 8.48 \times 10^{-10} \text{ m}$$

nth Orbits radius:-

$$r_n = n^2 \times 5.3 \times 10^{-11} \text{ m}$$

$$= 1.325 \times 10^{-9} \text{ m}$$

Calculate the radius of the third orbit of hydrogen atom:-

⇒ we know:-

$$r_n = n^2 \times 5.3 \times 10^{-11} \text{ m}$$

$$\therefore r_3 = 3^2 \times 5.3 \times 10^{-11} \text{ m}$$

$$= 4.77 \times 10^{-10} \text{ m}$$

Ans:-

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Energy of Electron in each Orbit

ই-র প্রতি কক্ষের শক্তি-স্বরূপ

⇒ For hydrogen atom, the total Energy of the electron, is the sum of its kinetic energy (KE) = $\frac{1}{2}mv^2$ and potential energy (P.E) = $-\frac{e^2}{r}$.

$$\therefore E = \frac{1}{2}mv^2 + \left(-\frac{e^2}{r}\right)$$

$$= \frac{1}{2}mv^2 - \frac{e^2}{r} \quad \text{--- (i)}$$

we know:-

$$\frac{e^2}{r} = mv^2$$

Substituting the value of mv^2 in eq (i) =

$$\therefore E = \frac{1}{2} \times \frac{e^2}{r} - \frac{e^2}{r}$$

$$= -\frac{1}{2} \times \frac{e^2}{r} \quad \text{--- (ii)}$$

Again we know:-

$$r = \frac{n^2 h^2}{4\pi^2 m}$$

Substituting the value of r in eq (ii) =

$$\therefore E = -\frac{1}{2} \times e^2 \times \frac{4\pi^2 m}{n^2 h^2}$$

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

$$\Rightarrow E = - \frac{2\pi^2 m e^4}{n^2 h^2}$$

$$\therefore E_n = - \frac{1}{n^2} \times 2.18 \times 10^{-18} \text{ J/atom}$$

$n = \text{no. of orbit.}$

$$\left\{ \begin{array}{l} \pi = 3.1416 \\ e = 1.6 \times 10^{-19} \\ h = 6.626 \times 10^{-34} \\ m = 9.11 \times 10^{-31} \\ E_0 = 8.85 \times 10^{-12} \end{array} \right.$$

#problem 5:-

$\Rightarrow$ Calculate the five lowest energy levels of the hydrogen atom.

$\Rightarrow$ we know:-

$$E_n = - \frac{1}{n^2} \times 2.18 \times 10^{-18} \text{ J/mol}$$

For 1st lowest Energy levels:-

$$n = 1$$

$$\therefore E_1 = - \frac{1}{1^2} \times 2.18 \times 10^{-18} \text{ J/mol}$$

$$= -2.18 \times 10^{-18} \text{ J/mol}$$

For 2nd lowest Energy levels:-

$$n = 2$$

$$E_2 = - \frac{1}{4} \times 2.18 \times 10^{-18} \text{ J/mol}$$

$$= -5.45 \times 10^{-19} \text{ J/mol}$$

For 3rd lowest Energy :-

$$n=3$$

$$\therefore E_3 = -\frac{1}{9} \times 2.18 \times 10^{-18} \text{ J/mol}$$

$$= -2.422 \times 10^{-19} \text{ J/mol}^{-1}$$

For 4th lowest Energy :-

$$E_4 = -\frac{1}{16} \times 2.18 \times 10^{-18} \text{ J/mol}^{-1}$$

$$= -1.362 \times 10^{-19} \text{ J/mol}^{-1}$$

For 5th lowest Energy :-

$$E_5 = -\frac{1}{25} \times 2.18 \times 10^{-18} \text{ J/mol}^{-1}$$

$$= -8.72 \times 10^{-20} \text{ J/mol}^{-1}$$

#problem 6 :-

⇒ Calculate the energy of 2nd orbit of the hydrogen atom.

⇒ we know —

$$E_n = -\frac{1}{n^2} \times 2.18 \times 10^{-18} \text{ J/mol}$$

$$\therefore E_2 = -\frac{1}{4} \times 2.18 \times 10^{-18} \text{ J/mol}^{-1}$$

$$= -5.45 \times 10^{-19} \text{ J/mol}^{-1}$$

Ans :-

Values of Rydberg's Constant:-

We know:-

the Energy of the electron in orbit n . is

$$E_n = - \frac{2\pi^2 e^4 m}{n^2 h^2}$$

∴ For n_1 :-

$$E_1 = - \frac{2\pi^2 e^4 m}{n_1^2 h^2}$$

For n_2 :-

$$E_2 = - \frac{2\pi^2 e^4 m}{n_2^2 h^2}$$

∴ Now the difference of Energy:-

$$\begin{aligned} \Delta E = E_2 - E_1 &= - \frac{2\pi^2 e^4 m}{h^2} \left(+ \frac{1}{n_2^2} - \frac{1}{n_1^2} \right) \\ &= \frac{2\pi^2 e^4 m}{h^2} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \end{aligned} \quad (1)$$

According to Planck's Equation:-

$$\Delta E = h\nu = \frac{hc}{\lambda}$$

Substituting the value of ΔE in the eq (1) =

$$\begin{aligned} \frac{hc}{\lambda} &= \frac{2\pi^2 e^4 m}{h^2} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \\ \therefore \frac{1}{\lambda} &= \frac{2\pi^2 e^4 m}{h^3 c} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \end{aligned}$$

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

$$\text{or } \boxed{\frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]}$$

$$R = \text{Rydberg's Constant} = \frac{2\pi^2 e^4 m}{h^3 c}$$

$$R = 10967800 \text{ m}^{-1}$$

problem - 21

⇒ Calculate the wave length in Å of the line in Balmer Series that is associated with drop of the electron from the fourth Orbit. the value of Rydberg Constant is 109676 cm^{-1}

⇒ Is given:

$$n_1 = 2 \rightarrow \text{Since Balmer Series.}$$

$$n_2 = 4$$

$$R = 10967600 \text{ m}^{-1}$$

we know:

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\frac{1}{\lambda} = 10967600 \left(\frac{1}{2^2} - \frac{1}{4^2} \right)$$

$$\therefore \lambda =$$

$$4862.8 \times 10^{-10}$$

$$= 4862.8 \text{ Å}$$

Ans.

problem - 8! —

⇒ Calculate the wave-length in Å of the third line in Balmer series that is associated with drop of the electron. $R = 109676 \text{ cm}^{-1}$

⇒ In given! —

$n_1 = 2 \rightarrow$ for Balmer series.

$$n_2 = 3$$

$$R = 109676 \text{ cm}^{-1}$$

we know! —

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\therefore \lambda = \left(R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \right)^{-1}$$

$$= \left(109676 \times \left(\frac{1}{2^2} - \frac{1}{3^2} \right) \right)^{-1}$$

$$= 6564.70 \text{ Å}$$

Ans! —

problem - 9! —

⇒ Calculate the wave length of the the emitted radiation when the e^- drops 4th to 1st orbit.

we know! —

$$\lambda = \left(R \times \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \right)^{-1}$$

$$= \left(109678 \times \left(\frac{1}{1^2} - \frac{1}{4^2} \right) \right)^{-1}$$

$$= 072.54 \text{ Å}$$

In given! —

$$n_1 = 1$$

$$n_2 = 4$$

$$R = 109678 \text{ cm}^{-1}$$

Ans! —

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Wave Mechanical Concept of the Atom

Ch: 02

#Wave Mechanical Theory:-

Q:- What does the wave mechanical theory state.

⇒ The wave mechanical theory states that the energy of electron is quantised and that this energy increases with increasing quantum numbers.

#Dual Nature of Matter:-

Q:- What is de Broglie dual Nature of matter:-?

⇒ He proposed that as light exhibits both wave-like and particle-like properties, matter also exhibits particle-like and wave-like properties.

(ଜିନିଷ ଦୁଇଟି ଧର୍ମ କରେ । ଗୋଟିଏ ହେଉଛି ତରଙ୍ଗ-ସଦୃଶ ଏବଂ ଅନ୍ୟଟି କଣିକା-ସଦୃଶ ଧର୍ମ କରେ । ଗୋଟିଏ ପାର୍ଟିକିଲ ତରଙ୍ଗ-ସଦୃଶ ଏବଂ ଅନ୍ୟଟି ତରଙ୍ଗ-ସଦୃଶ ଧର୍ମ କରେ ।)



Heisenberg's Uncertainty Principle:-

(स्थिति और गति का अनिश्चितता सिद्धांत)

⇒ Heisenberg's Uncertainty principle state that it is impossible to measure or calculate exactly both the position and the momentum of an object.

1. The position and momentum of a moving particle are interdependent and cannot be measure with absolute exactness.

2. If momentum is measured accurately, the position become uncertain and less accurate; If the position is measured accurately, the momentum become uncertain and less accurate.

Heisenberg's Relationship

Can be shown as:-

$$\Delta x \times \Delta p \geq \frac{h}{4\pi}$$

$$\Rightarrow \Delta x \times m \Delta v \geq \frac{h}{4\pi}$$

$$\Rightarrow \boxed{\Delta x \times \Delta v \geq \frac{h}{4\pi m}}$$

Δx = Error In position measurement

Δp = Error In momentum measurement

Δv = Error In Velocity measurement.

De Broglie's Equation:-

⇒ Apply the de Broglie wave equation $\lambda = \frac{h}{mc}$ to solve for the wavelength of the moving electron.

* De Broglie's Equation Derivation

→ According to plank quantum theory:-

we know:-

$$E = h\nu \dots \dots \textcircled{1}$$

→ Einstein theory of relativity:-

we know:-

$$E = mc^2 \dots \dots \textcircled{2}$$

From eq (1) & (2) ⇒

$$mc^2 = h\nu$$

$$\Rightarrow mc^2 = \frac{hc}{\lambda} \quad \left[\because \nu = \frac{c}{\lambda} \right]$$

$$\Rightarrow mc = \frac{h}{\lambda}$$

$$\therefore \boxed{\lambda = \frac{h}{mc}}$$

$$\lambda = \frac{h}{\text{momentum}}$$

$$\lambda \propto \frac{1}{\text{momentum}}$$

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Where:-

E = photon energy.

h = planck constant

ν = frequency

c = velocity

m = mass of photon.

Note:- The momentum of a particle in motion is inversely proportional to wavelength.

1. This equation is applicable to all particles, but the wave like aspect is typically significant for very small particles such as electrons. (ଏହା ସମସ୍ତ କଣিকା ପାଇଁ ଉପଯୋଗୀ, କିନ୍ତୁ ତରଙ୍ଗ ପରିଚ୍ଛନ୍ନତା କେବଳ କ୍ଷୁଦ୍ର କଣିକା ଯେପରିକି ଇଲେକ୍ଟ୍ରନ୍ ପାଇଁ ଉଲ୍ଲେଖନୀୟ)

2. Large particles, such as 100g stone with 1000 cm/s velocity, have practically unmeasurable wavelengths. (6.62×10^{-34} cm)

3. For a small masses like atom, the wavelength can be detectable.

Ex:- Hydrogen mass = 1.6×10^{-28} g

" velocity = 2.18×10^8 cm/s

$$\therefore \lambda = \frac{h}{mv} = 3.32 \times 10^{-8} \text{ cm}$$

this value comparable to "X-rays"

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

problem: 1: _____

⇒ Calculate the wavelength of an α -particle having mass $6.6 \times 10^{-27} \text{ kg}$ moving with a speed of 10^5 cm sec^{-1}
($h = 6.6 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$)

⇒ In given: _____

$$\text{mass } m = 6.6 \times 10^{-27} \text{ kg}$$

$$v = 10^5 \text{ cm s}^{-1} = 10^5 \times 10^{-2} \text{ m s}^{-1}$$

we know:

$$\lambda = \frac{h}{mv}$$

$$= \frac{6.6 \times 10^{-34}}{6.6 \times 10^{-27} \times 10^5 \times 10^{-2}}$$

$$= 1 \times 10^{-10} \text{ m}$$

problem: 2: _____

⇒ Calculate the uncertainty in position of an electron if the uncertainty in velocity is $5.7 \times 10^5 \text{ m s}^{-1}$,

⇒ In given: _____

$$\Delta v = 5.7 \times 10^5 \text{ m s}^{-1}$$

$$\text{and mass of } e^- m = 9.1 \times 10^{-31} \text{ kg} \quad \left[h = 6.626 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1} \right]$$

we know:

$$\Delta x \times \Delta v = \frac{h}{4\pi m}$$

$$\therefore \Delta x = \frac{h}{4\pi m \Delta v}$$

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

$$\therefore \Delta x = \frac{6.626 \times 10^{-34}}{4\pi \times 0.1 \times 10^{-9} \times 5.7 \times 10^5}$$

$$= 1.012 \times 10^{-10} \text{ m}$$

Ans:-

Schrödinger's Wave Equation:-

⇒ A mathematical expression describing the energy and position of the electron in space and time.

$$\frac{d^2\psi}{dx^2} + \frac{d^2\psi}{dy^2} + \frac{d^2\psi}{dz^2} + \frac{8\pi^2m}{h^2}(E - P.E)\psi = 0$$

where,

ψ = wave motion

x, y, z = distance from nucleus in three axis.

λ = wavelength

A = Constant (max value of ψ)

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Probability Distribution Curves:-

⇒ Radial distribution curve gives an idea about the e^- density at a radial distance from the nucleus.

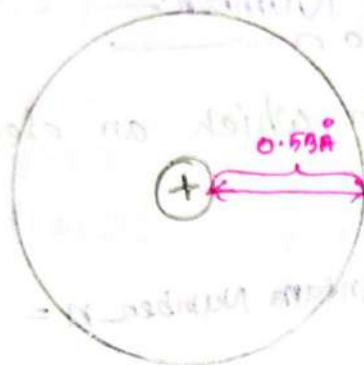


Fig:- the probability distribution of e^- cloud

Quantum Numbers: (Q.N)

⇒ The set of numbers used to describe the position and energy of the electron in an atom, are called Quantum Numbers. (ଏକାନ୍ତ ସଂଖ୍ୟା ଯେ ଇଲେକ୍ଟ୍ରନ୍ ସହଜରେ କଣିକା ଭାବରେ ବର୍ଣ୍ଣନା କରାଯାଇପାରେ ତାହାକୁ କୁହାଯାଏ।)

There are 4 types of Quantum Numbers:-

1. Principle Quantum Number (n); (ମୁଖ୍ୟ କ୍ୱାଣ୍ଟମ ସଂଖ୍ୟା)
2. Subsidiary Quantum Number (l); (ସାହାଯ୍ୟ କ୍ୱାଣ୍ଟମ ସଂଖ୍ୟା)

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

3. Magnetic Quantum Number (m); ଅଧ୍ୟାକାଶ (କୋଅର୍ଡିନେଟ) ସଂଖ୍ୟା

4. Spin Quantum Number (s); ସ୍ପିନ୍ (କୋଅର୍ଡିନେଟ) ସଂଖ୍ୟା

Principal Quantum Number:-

⇒ The main shell in which an electron travels around the nucleus.

principle Quantum Number $n = 1 \dots 2 \quad 3 \quad 4$

Letter designation $K \quad L \quad M \quad N$

Max e^- No.

$2n^2 = 2 \quad 8 \quad 18 \quad 32$

Subsidiary Quantum Number:-

⇒ A Quantum Number that determines its orbital Angular momentum.

For any given value of the principal quantum ' n ', the Subsidiary Quantum Number ' l ' have All integral values from $\{0 \text{ to } (n-1)\}$.

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Ex:- For 3rd Energy level:-

$$n=3$$

$$l=0, 1, 2$$

For 4th Energy level:-

$$n=4$$

$$l=0, 1, 2, 3$$

$$\Rightarrow n=1,$$

$$l=0(1s)$$

$$n=2,$$

$$l=0(2s)$$

$$l=1(2p)$$

$$n=3,$$

$$l=0(3s)$$

$$l=1(3p)$$

$$l=2(3d)$$

$$n=4,$$

$$l=0(4s)$$

$$l=1(4p)$$

$$l=2(4d)$$

$$l=3(4f)$$

Magnetic Quantum Number:-

$\Rightarrow$ That state the position of the electron with respect to the nucleus.

For any given value of the subsidiary Quantum Number 'l', the magnetic Quantum Number 'm' have All Integral Values from $\{+l \text{ to } -l\}$.

$$\therefore \text{total values of } m = 2l + 1$$

Ex:- $l=2$

$$m = -2, -1, 0, 1, 2$$

$$l=2$$

$$m = \pm 2, \pm 1, 0$$

TOPIC NAME : _____

DAY : _____

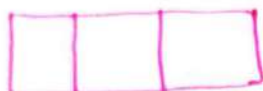
TIME : _____

DATE : / /

Degenerate Orbitals:

⇒ Orbitals having same energy are known as degenerate orbital.

Ex: 1. p orbital has 3 same energy level degenerate orbital.



p_x p_y p_z

that said to be three fold degenerate orbital.

2. d orbital have 5 same energy level degenerate orbital. that said to be five fold degenerate orbital.

Node & nodal plane:

⇒ The nucleus at which the electron density is zero is called node and plane passing through the orbital axis is called nodal plane.

Lobe:

⇒ The shape of e^- waves and refers to a high probability density area to finding an electron.

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Shape of different

different Subshell:-

P Subshell shapes:-

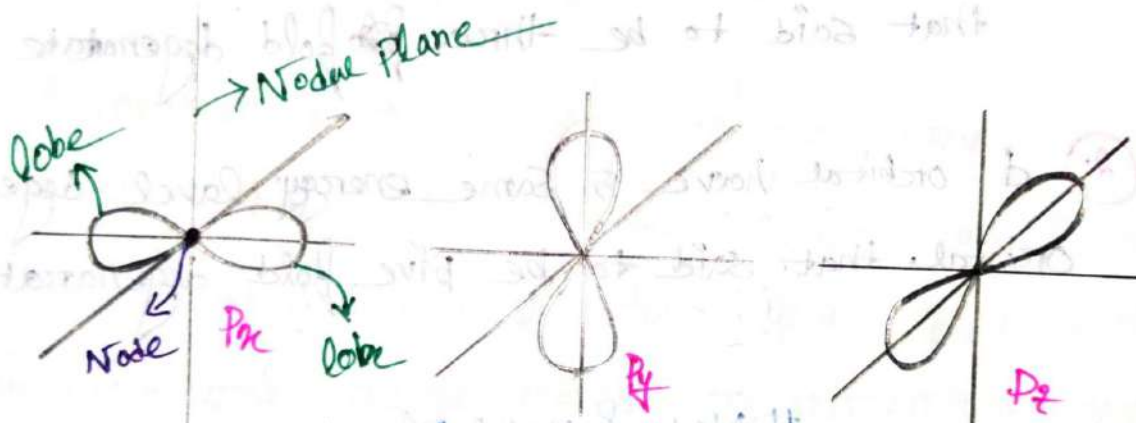
For p Subshell:-

$$n = 2$$

$$l = 1$$

$$m = -1, 0, 1$$

$$s = \pm \frac{1}{2}$$



d Subshell shapes:-

For d Subshells:-

$$n = 3$$

$$l = 2$$

$$m = \pm 2, \pm 1, 0$$

$$s = \pm \frac{1}{2}$$

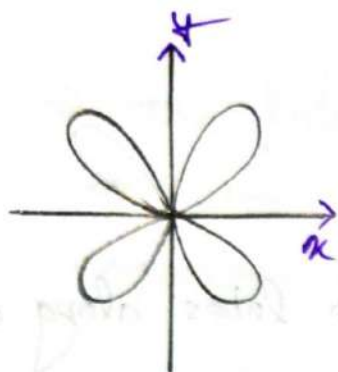
d orbitals have 5 small suborbitals they are,
 $d_{xy}, d_{yz}, d_{zx}, d_{x^2-y^2}, d_{z^2}$

TOPIC NAME : _____

DAY : _____

TIME : _____

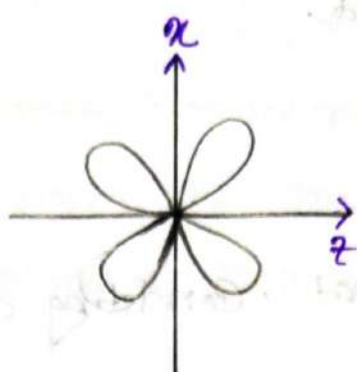
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d_{xy}

$l=2, m=-2$

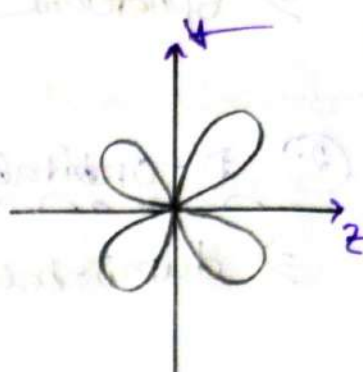
double dumb-bell
Shaped (D.D.B.S)



d_{zx}

$l=2, m=+1$

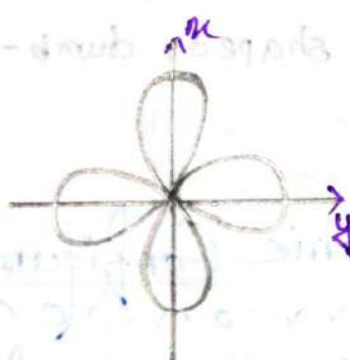
(D.D.B.S)



d_{yz}

$l=2, m=-1$

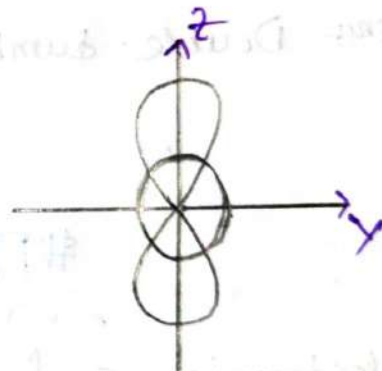
(D.D.B.S)



$dx^2 - y^2$

$l=2, m=+2$

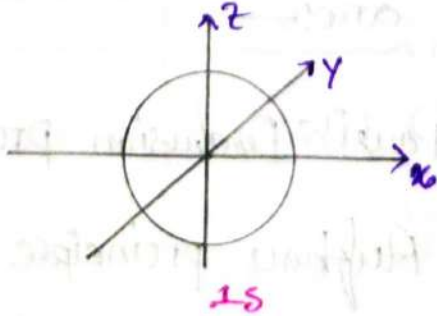
5 orbitals



dz^2

$l=2, m=0$

(D.B.S)



1s

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

① s-orbitals:-

⇒ spherical shaped.

② p-orbitals:-

⇒ dumb-bell shaped, consisting of two lobes along axis.

③ d-orbitals:-

⇒ Complex-geometrical shapes, lobes lie in the plane.
(ex: Double-dumb-bell shaped, dumb-bell shaped)

#Electronic Configuration:-

⇒ electronic configuration of atom follows the three principles:-

there are:-

1. Pauli's Exclusion principle.

2. Aufbau principle.

3. Hund's Rule:-

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Aufbau principle:-

⇒ electrons fill lower-energy atomic Orbitals before filling higher-energy atomic Orbitals. (ପ୍ରଥମେ e^- ନିମ୍ନ-ଶକ୍ତିର ଅଟମିକ ଅରବିଟାଲ୍ ଗୁଡ଼ିକରେ ଉପସ୍ଥାନ କରେ ଏବଂ ତା'ପରେ ଉଚ୍ଚ-ଶକ୍ତିର ଅଟମିକ ଅରବିଟାଲ୍ ଗୁଡ଼ିକରେ ଉପସ୍ଥାନ କରେ)
(ଅର୍ଥାତ୍ - $1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d, 7p$)

Comparison of e^- energies in different Orbitals:-

⇒ The energy of an orbital is determined by sum of ' n ' and ' l '. this is called (n+l) rule.

Note:- The more the value of (n+l) for an orbital, the higher its energy. (ଯଦି ଅଟମିକ ଅରବିଟାଲ୍ (n+l) ର ମାନ ଅଧିକ, ତେବେ ଏହାର ଶକ୍ତି ଅଧିକ ହେବ।)

Q:- which has more energy between $3s$ and $3p$.

We know:-

for $3s$:-

$$n = 3$$

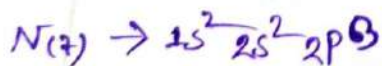
$$l = 0$$

$$\therefore n + l = 3 + 0 = 3$$

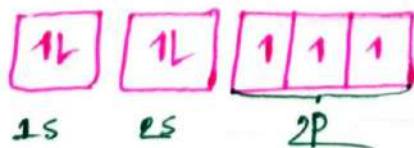
$$\therefore n+l = 3+1 = 4$$

that means 3p has more/higher energy compare to 3s.

(1) $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$



According to Hund's Rule:—

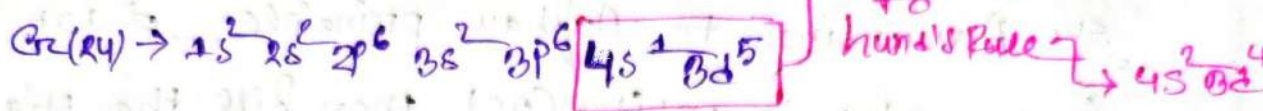
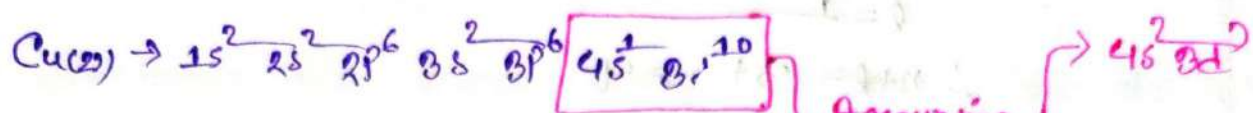


TOPIC NAME : _____

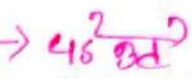
DAY : _____

TIME : _____ DATE : / /

Some Exception of Hund's Rule:-



According to Hund's Rule



$\Rightarrow$ This is happened because for a more stable configuration half-filled and full-filled are preferred.

If (n+l) values are equal:-

Q:- which has more energy between 3d and 4p?

$\Rightarrow$ According to (n+l) rules:-

For 3d:-

$n = 3$

$l = 2$

$\therefore (n+l) = 3+2 = 5$

$$\left. \begin{array}{l} l=0, s \\ l=1, p \\ l=2, d \\ l=3, f \end{array} \right\}$$

For 4p:-

$n = 4$

$l = 1$

$\therefore (n+l) = 4+1 = 5$

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

In this case, which orbital has smaller principle quantum Number has lower energy. (ନିମ୍ନେ ଉଲ୍ଲେଖ କରାଯାଇଥିବା ଅଣୁର ଓର୍ବିଟାଲ୍ ସମସ୍ତଙ୍କ ମଧ୍ୟରୁ କେଉଁ ଓର୍ବିଟାଲ୍ କର୍କମ ଶକ୍ତି ବିଶିଷ୍ଟ ଅଟେ)

So, $4d < 4p$

problems 10: _____

⇒ What is the wave-length associated with a particle of mass $0.1g$ moving with a speed of $1 \times 10^6 \text{ cm s}^{-1}$. $h = 6.6 \times 10^{-27} \text{ erg sec}$.

⇒ Is given: _____

$$m = 0.1g = 0.1 \times 10^{-3} \text{ kg}$$

$$c = 1 \times 10^6 \text{ cm s}^{-1} = 1 \times 10^5 \times 10^{-2} \text{ m}$$

$$h = 6.6 \times 10^{-27} \text{ erg sec} = 6.6 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$$

we know: _____

$$\lambda = \frac{h}{mc}$$

$$= \frac{6.6 \times 10^{-34}}{1 \times 10^5 \times 10^{-2} \times 0.1 \times 10^{-3}}$$

$$= 6.6 \times 10^{-33} \text{ m}$$

Ans:—

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

problems 11: _____

⇒ The uncertainty in the position of a moving bullet of mass 0.01 kg is $1.0 \times 10^{-5} \text{ m}$. Calculate the uncertainty in its velocity:—

⇒ Is given:—

$$\Delta x = 1.0 \times 10^{-5} \text{ m}$$

$$m = 0.01 \text{ kg}$$

$$h = 6.626 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$$

we know:—

$$\Delta x \times \Delta v = \frac{h}{4\pi m}$$

$$\therefore \Delta v = \frac{h}{\Delta x \times 4\pi m}$$

$$= \frac{6.626 \times 10^{-34}}{1.0 \times 10^{-5} \times 4 \times \pi \times 0.01}$$

$$= 5.272 \times 10^{-28} \text{ m s}^{-1}$$

Ans:—

problems 12:— What is the mass of a photon of Sodium light with a wavelength of 5800 Å .

⇒ Is given:—

$$\lambda = 5800 \text{ Å} = 5800 \times 10^{-10} \text{ m}$$

$$h = 6.626 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$$

$$c = 3 \times 10^8 \text{ m s}^{-1}$$

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

we know:-

$$\lambda = \frac{h}{mc}$$

$$\therefore m = \frac{h}{\lambda c}$$

$$= \frac{6.626 \times 10^{-34}}{5800 \times 10^{10} \times 3 \times 10^8}$$

$$= 3.790 \times 10^{-36} \text{ kg}$$

$$= 3.740 \times 10^{-36} \text{ kg}$$

Ans:-

problem: 18:-

⇒ The uncertainty in the position and velocity of a particle are 10^{-10} m and $5.27 \times 10^{-24} \text{ m s}^{-1}$ respectively. Calculate the mass of the particles. where $h = 6.6 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$

⇒ Is given:-

$$\Delta x = 10^{-10} \text{ m}$$

$$\Delta V = 5.27 \times 10^{-24} \text{ m s}^{-1}$$

$$h = 6.6 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$$

we know:-

$$\Delta x \times \Delta V = \frac{h}{4\pi m}$$

$$\therefore m = \frac{h}{\Delta x \Delta V \times 4\pi}$$

$$= \frac{6.6 \times 10^{-34}}{1 \times 10^{-10} \times 5.27 \times 10^{-24} \times 4\pi} = 0.000 \text{ kg}$$

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

#problem: 14:-

⇒ The velocity of a ball being bowled by M.D. Rafiq is 25 m s^{-1} . Calculate the wavelength of the matter-wave associated with ball. (Weight of the ball = 158.5 g $h = 6.625 \times 10^{-34} \text{ erg sec}$)

⇒ Is given:-

$$v = 25 \text{ m s}^{-1}$$

$$h = 6.625 \times 10^{-34} \text{ erg sec}$$

$$= 6.625 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$$

$$m = 158.5 \text{ g} = 158.5 \times 10^{-3} \text{ kg}$$

We know:-

$$\lambda = \frac{h}{mv}$$

$$= \frac{6.625 \times 10^{-34}}{158.5 \times 10^{-3} \times 25}$$

$$= 1.67 \times 10^{-34} \text{ kg}$$

#problem: 15:-

(a) An electron is the $4f$ orbital. What possible values for the quantum numbers n, l, m, s can it have.

⇒ Is given:-

orbital = $4f$

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

So, For 4f orbital:-

$$n = 4$$

$$l = 3$$

$$m = \pm 3, \pm 2, \pm 1, 0$$

$$s = \pm \frac{1}{2}$$

Ans:-

(b) What designation is given to an orbital having

(i) $n = 2, l = 1$

(ii) $n = 4, l = 0$

(i)

Is given:-

principle Quantum Number $n = 2$

Subsidiary Quantum Number $l = 1$

$l = 1$ corresponds to a p orbital

$\therefore$ the designation for this orbital is = 2p

(ii)

Is given:-

$n = 4$ which corresponds to the 4th energy level

$l = 0$ which corresponds to the 's' orbital.

$\therefore$ the designation for this orbital is = 4s.

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

#problem 16! —

⇒ A neutral atom has $2K, 8L, 5M$ electrons. Find out the following from the data.

- (a) atomic Number.
- (b) total Number of 's' electrons.
- (c) total Number of 'p' electrons.
- (d) number of protons in the nucleus.
- (e) Valency of elements.

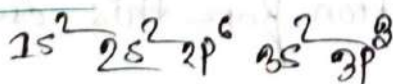
⇒ In given! —

The e^- configuration

$2K, 8L, 5M$

(a)

so, the e^- will be —



The atomic Number is the total Number of e^- in a Neutral atom.

$$\therefore Z = 2 + 8 + 5 = 15$$

TOPIC NAME : _____

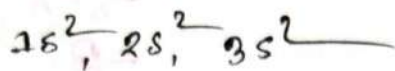
DAY : _____

TIME : _____

DATE : / /

(b)

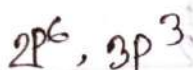
In the e^- configuration, s orbital are:-



∴ the the total No. of s electron = $2+2+2$
 $= 6e^-$

(c)

In the e^- configuration p orbital are:-



∴ the total No. of 'p' electron = $6+3$
 $= 9$

(d)

Since the Atom is Neutral

∴ the Number of protons = No. of e^- = No. of atomic no.

∴ No. of protons = 15.

(e)

~~Since~~ The Valency is related to the number of e^- in the outmost shell. For this atom, the outmost shell is the 'n' shell with $5e^-$.

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

$\therefore$ The valency is given by $= 8 - \text{no. of } e^- \text{ in the outmost shell}$

$$= 8 - 5$$

$$= 3$$

Ans:-

problems 12:-

$\Rightarrow$ State which of the following sets of quantum numbers would be possible, and which would be impossible.

(a) $n=0, l=0, m=0, s=+\frac{1}{2}$

$\Rightarrow$ that is impossible. Because of primary quantum no can't be negative or 0. It must be positive

(b) $n=1, l=1, m=0, s=+\frac{1}{2}$

$\Rightarrow$ that is impossible, because of for $n=1$, l can only be 0. so $l=1$ is not allowed.

(c) $n=1, l=0, m=0, s=-\frac{1}{2}$

$\Rightarrow$ It's possible, cause this satisfies the rules of quantum number.

(d) $n=2, l=1, m=+2, s=+\frac{1}{2}$

$\Rightarrow$ that is not possible. Because of the value of m can't be greater than the value of l .

②. $n=2, l=1, m=-1, s=+\frac{1}{2}$

⇒ It's possible, Because this set satisfies the rule of quantum numbers.

problems 18: —

① (i)

⇒ How many electrons are there in hydrogen and chlorine atom (atomic no. 17)? How they are arranged?

② (ii) What is the valency of hydrogen and chlorine in HCl .

⇒ we know: —

①

H has an atomic Number $Z=1$

∴ the e^- of H = $Z=1$

Cl has an atomic Number $Z=17$

∴ the e^- of Cl = $Z=17$

② (ii)

⇒ we know: —

The Valency is related to the no. of e^- in the outmost shell.

For H atom: —

outmost shell is K with $1e^-$

∴ Valency = $2-1=1$

For Cl atom: —

outmost shell is 'm' with $7e^-$

∴ Valency = $8-7=1$

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

So, in HCl, hydrogen has a valency of 1, and chlorine has a valency of 1. The e^- sharing between H and Cl results in the formation of HCl molecule.

⑥

⇒ The atomic number of Na and Cl are 11 and 17 respectively. Determine the no of e^- in Na^+ and Cl^- .

⇒ Is given:-

The atomic Number of Na = 11

∴ the e^- of Na = 11 in Neutral state.

Now:- For Na^+ ion, it lose an e^- to achieve an stable electron configuration similar to (Ne) ($1s^2 2s^2 2p^6$).

∴ Na^+ ion has 10 e^- .

The atomic Number of Cl = 17

∴ in Neutral state e^- No. of Cl = atomic no.
= 17

Now:- For Cl^- ion, it gains an e^- to achieve an stable electron configuration similar to Argon (Ar = $1s^2 2s^2 2p^6 3s^2 3p^6$).

∴ Cl^- ion has = 18 e^-

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

MODERN PERIODIC TABLE

Ch:- 03

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Periodic table:-

⇒ The periodic table is a tabular array of the chemical elements organized by atomic number.

Modern periodicity:-

⇒ The modern periodic law states that when elements are arranged in order of increasing atomic number, there is periodic repetition of their chemical and physical properties.

The Cause of Periodicity:-

⇒ The cause of periodicity in properties is the repetition of similar outer electronic configuration at certain regular intervals.

Main Features of Modern Periodic Table:-

⇒ Modern periodic table is based on the electronic configuration.

1. It consists 18 vertical columns (Groups)

2. and 7 horizontal rows (Periods) (अवधि)

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Groups:

⇒ Elements with similar properties are organized into columns called groups.

① Each group has elements with the same valence electron configuration but different principle quantum number.

② IUPAC number the groups from 1 to 18 in 1988.

③ groups are divided into two subgroups: A & B.

④ The VIIIA(18) group is also known as noble gases.

Properties of Groups

(प्रकार - विशेषताएं):

① Group 1: known as Alkali metals (प्रकार 1), highly reactive, have a +1 charge.

② Group 2: known as Alkali earth metals (प्रकार 2), have +2 charge.

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

③ Group 9-12 : Transition metal (very reactive) . Found in the middle of the periodic table.

④ Lanthanides & Actinides : Inner transition metal (very reactive) located at the bottom.

⑤ Group 17 : Halogens group, highly reactive, have a -1 charge.

⑥ Group 18 : Noble Gases / Inert Gases, chemically inert.

⑦ Note :

⇒ Hydrogen : Occupies a unique position, does not fit naturally into any group, has a single positive charge.

⇒ Group VIII B (8, 9, 10) : Contains triads of elements with similar properties, typically metals.

#periods (rows)

⇒ The Horizontal rows of the periodic table are called periods.

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Properties of Periods:-

(ପ୍ରତିକ୍ରମ ବିଶେଷତା)

- ① The 1st period has two elements, H, & He.
- ② The 2nd and 3rd periods contain 8 elements each.
- ③ The 4th and 5th periods consist of 18 elements each.
- ④ The 6th and 7th periods consist of 32 elements each.
- ⑤ A separate panel at the bottom includes the Lanthanides (16 elements of the 6th) and Actinides (16 elements of the 7th period).
- ⑥ Elements 113, 115, 117, 118 were confirmed in 2016. named nihonium (Nh), moscovium (Mc), tennessine (Ts), oganesson (Og).
- ⑦ Lanthanides (La-Lu) and actinoids (Ac-Lr) are collective name recommended by IUPAC.

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Classification of elements

Into Blocks:—

S-Block elements:—

⇒ The s-Block element have valance configuration s^1 or s^2

general e⁻ configuration → ns^{1-2}

groups → 1 - 2

periods → 1 - 7

No. of element → 14

P-Block elements:—

⇒ The p-Block element have valance configuration s^2p^1 to s^2p^6

general e⁻ configuration → $ns^2 np^{1-6}$

groups → 13 - 18

periods → 2 - 7

No. of elements → 36

d-Block elements:—

⇒ Elements whose last e⁻ enters the d orbital are called d-Block element. And have valance configuration in which d-subshell are being filled.

general e⁻ configuration → $ns^{1-2} (n-1)d^{1-10}$

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

groups $\rightarrow 3-12$

periods $\rightarrow 4-7$

No. of elements $\rightarrow 41$

f-Block elements:

$\Rightarrow$ Elements whose last electron enters the f orbital are called f-block element. And have valence configuration in which f-subshell are being filled.

general e configuration $\rightarrow ns^2 (n-1)d^{0-2} (n-2)f^{1-14}$

groups $\rightarrow 3$

periods $\rightarrow 6-7$

No. of elements $\rightarrow 28$

Classification of elements depending on e configuration:

1. ^{$\rightarrow$ Inert gases} The Inert Gases (elements of 's' group)
2. ^{$\rightarrow$ Representative} The representative element ('s' and 'p' block elements)
3. ^{$\rightarrow$ Transition} The Transition element ('d' block element)
4. ^{$\rightarrow$ Inner transition} The Inner Transition element ('f' block element)

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Metals :-

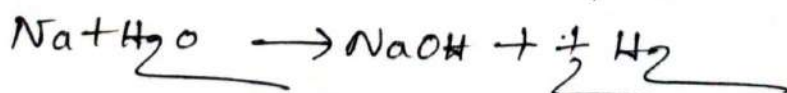
⇒ Most of the elements in the periodic table are metals. They are divided into groups. Like, alkali metal, alkali earth metal, transition metal, inner transition metals.

Properties of metals :-

1. Lustrous (Shiny) Appearance (उज्ज्वल (चमकता) + रंग)
2. Malleable & (capable of being hammered into sheets).
3. Ductile : (can be drawn into wire)
4. Good conductors of heat & electricity.
5. Metallic elements, **except mercury (Hg)**, are typically solids at room temperature (20°C).
6. Metallic properties stem from high mobility of electrons in the outer shell of metal atoms.

Alkali Metals (-ग्रो १):-

⇒ Group 1 elements are called Alkali metals. They react with water to form strong alkalis.



TOPIC NAME: _____

The alkali metals are:- $\rightarrow$ Li, Na, K, Rb, Cs, FrProperties of Alkali metals:-

1. Hydrogen is placed in Group 1 but it's not metal.
2. Na and K are the 7th and 8th most abundant element in earth crust.
3. Highly reactive.
4. have $+1$ charge.
5. have the largest atom sizes in periods.

Uses of Alkali metals:-

1. Alkali metal compounds are common in nature and daily life.
2. NaCl (table salt)
3. Lithium compounds used in greases, batteries, and drugs.

TOPIC NAME : _____

DAY : _____

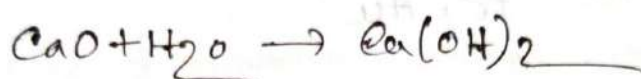
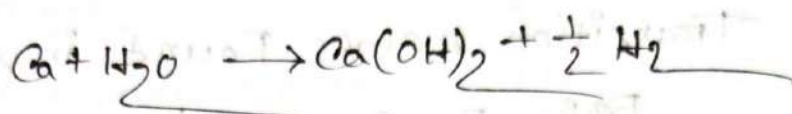
TIME : _____

DATE : / /

Alkaline Earth Metals:-

(Grp 2, 12)

⇒ Group 2 elements are called Alkaline Earth metals. they react with water to form Alkali.



Properties of Alkaline Earth Metals

- ⇒ ① Alkaline Earth Metals: Be, Mg, Ca, Sr, Ba, Ra.
- ② Beryllium, Strontium, and barium are rare.
- ③ Radium is highly radioactive.
- ④ Ca and Mg are abundant, found in limestone minerals.
- ⑤ Ions with +2 charge
- ⑥ Atoms smaller than Alkali metals.

Transition Metals

(অবস্থান্তর ধাতু)

⇒ An element whose d orbital is partially (d^{1-9}) filled is called a Transition metal.

Transition metals Found in groups 3 to 12

Ex:- Fe, Au

Properties of Transition Metals:

1. Very hard with high melting and boiling points.

2. Good electrical conductor.

3. Very malleable.

4. Positive oxidation states, +2 and +3 (Fe^{2+} , Fe^{3+})

5. Effective catalytic agents. (প্রকৃতকালে প্রতিকার করে করে)

6. Form colored and complex compounds.
(বর্ণহীন দ্রব তৈরি করে করে)

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

#Metal Triads:-

⇒ A small group of transition metal, which are metals with very similar properties, usually found together.

☐ Iron Triad:-

1. Member : Iron, Cobalt and nickel.

2. Examples of transition metals with similar properties.

☐ Palladium Triad:-

1. MEMBER:- Ruthenium, rhodium, and palladium.

2. Position just under the Iron Triad in periodic table.

☐ Platinum Triad:-

1. MEMBER:- Osmium, Iridium, and platinum.

2. Position below the palladium triad.

Lanthanides and Actinides

Lanthanides

1. Located in a block below the main body of the Periodic table.
2. Silvery metals with tendency to tarnish easily.
3. Relatively soft with high melting and boiling points.
4. Reactive, forming various compounds.
5. Used in lamps, magnets, lasers, and for enhancing other metals.

Actinides

1. Located in a block below the Lanthanides row.
2. Atomic Number follow Actinium.
3. All actinides are radioactive, with positively charged ions.
4. Highly reactive metals forming compounds with nonmetals.
5. Used in medical/medicine and nuclear devices.
6. Have three incomplete outer levels (s, d, f orbitals).

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Nonmetals:

⇒ Non-metals are those which lack all the metallic attributes.
most of the non-metal are gases (e.g., O₂) or solids (P, S)

1. Solid non-metal are typically hard, lack metallic luster, and are brittle.

2. Bromine (Br) is the only liquid non-metal.

3. Includes halogens and noble gases.

Halogens:

⇒ Halogens are the group 17 of the table.

1. The Halogens element are: F, Cl, Br, I, At

2. React readily with metals to form salts (NaCl, CaCl₂)

3. Compounds with fluoride (F⁻) ions added to toothpaste for dental care.

4. Br, I, less abundant than Cl, At (Astatine) Highly radioactive and rare.

5. Found in bleaches, disinfectants.

6. Form ions with -1 charge.

7. Highly radioactive non-metals.

Noble Gases:-

⇒ Noble gases are the group 18/0 in the periodic table.

1. They are: He, Ne, Ar, Kr, Xe, Rn.

2. Monatomic gases composed of single atoms.

3. Unreactive at room temperature and pressure.

4. Previously called 'Inert gases' or 'rare gases' due to lack of reactivity.

5. He found in Natural gas.

5.1 Ar used to an inert atmosphere in welding and light bulbs.

6. Ne emits red light in gas discharge tubes, used in neon light.

Q:- what are noble gases? why are they called noble gases? And what are the uses of noble gases?

⇒ Noble gases are the group 18 in the periodic table. they are He, Ne, Ar, Kr, Xe, Rn. They are monatomic gases composed of single atoms, and unreactive under normal conditions.

They called Noble Gases:-

1. Due to lack of reactivity

2. The term "Noble" reflects their perceived inertness and stability.

Uses of Noble Gases:-

1. Helium (He):-

- (i) Inflating balloons.
- (ii) Helium-Neon Lasers.
- (iii) Aerospace Application.

2. Neon (Ne):-

- (i) Neon Lights and Signs.
- (ii) Advertising display.
- (iii) Indicator lamps.

3. Argon (Ar):-

- (i) For Inert Atmosphere.
- (ii) Semiconductor manufacturing.

4. Krypton (Kr) & Xenon (Xe):-

- (i) Flash Photography.

5. Radon (Rn):-

- (i) Radiotherapy for cancer treatment.
- (ii) Medical Research.

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Metalloids or Semimetals:-

⇒ A unique group of element's that share properties of both metal and nonmetals.

1. Elements are: B, Si, Ge, As, Sb, Te, Bi, Po

2. Enhance properties of both metals and Non-metals.

3. Good semiconductors, displaying increased electrical conductivity at higher Temp.

Periodic Table Trends:-

1. Atom size "decreases" left to right. "Increases" down a column.

(ପର୍ଯ୍ୟନ୍ତ ମାପନିତ ପଥ ଯେଉଁଠି ଯେଉଁ ଯେଉଁ ପରାମାପନ ଯାହା ଥାଏ
- ପଥ ଯେଉଁଠି ଯେଉଁ ମାପନିତ ଯେଉଁ ପରାମାପନ ଯାହା ହୁଏ
- ପଥ)

2. "Ionization" energy "Increases" left to right. "Decreases" down a column.

(ପର୍ଯ୍ୟନ୍ତ ମାପନିତ ପଥ ଯେଉଁଠି ଯେଉଁ ଯେଉଁ ଆୟନୀକରଣ ଶକ୍ତି ହୁଏ
- ପଥ ଯେଉଁଠି ଯେଉଁ ମାପନିତ ଯେଉଁ ଆୟନୀକରଣ ଶକ୍ତି ଥାଏ ନାହିଁ)

② Ability to form a chemical Bond "Increases" left to right, "Decreases" down a column.

(বাম থেকে ডান দিকে রাসায়নিক বন্ধন গঠনের ক্ষমতা বৃদ্ধি পায়, উপর থেকে নিচে দিকে গিয়ে দ্রুত হ্রাস পায়)।

③ Electron affinity decreases with increasing Atomic radius.

(পরমাণুর পারমাণবিক ব্যাসার্ধের সাথে e^- এর আকর্ষণ হ্রাস পায়)।

④ Electronegativity Increase left to right, Decreases down a column.

(ডান দিকে বৃদ্ধি পায় বাম থেকে ডান দিকে উপর থেকে নিচে দিকে হ্রাস পায়)।

Electronegativity

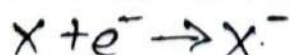
(-উচ্চ আকর্ষণ)

⇒ It measures how much an atom likes to attract e^- when forming bonds.

Electron Affinity

(e^- এর আকর্ষণ)

⇒ Energy released when an atom happily accepts an e^- .



TOPIC NAME : _____

Ionization Energy(Electron affinity)

⇒ Energy needed to pluck an e^- away from an atom.

# Usefulness of the periodicTable:-

1. classification of the element.
2. Prediction of undiscovered element.
3. Correction of Atomic weight.
4. Industrial research etc.

Limitations of thePeriodic Table:-

1. position of hydrogen.
2. position of Lanthanides and Actinides.
3. Non-Periodic properties.
4. Diagonal Relationship.

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Naturally Occurring and Synthetic Elements:

- ① 94 of 118 elements are naturally occurring.
- ② Technetium(Tc) and Promethium(Pm) are artificially prepared.
- ③ Trans-uranium elements (beyond uranium) are synthetic.
- ④ produced through nuclear transmutation reaction.

Element on Earth:-

Naturally Occurring Element:-

1. 94 elements found in Nature.
2. 12 elements make up 99.7% of Earth's Crust by mass.
3. Abundance (DESC ORDER BY) : O, Si, Al, Fe, Ca, Mg, Na, K, Ti, H, P, Mn.

Earth's Composition:-

1. Core predominantly Iron.
2. Surrounding mantle contains, Fe, O, Si, S.
3. Core-mantle structure at the center of Earth.

Elements in Human Body:-

⇒ The Essential Trace element in human Body:-

① Fe, Cu, Zn, I, Co

② Comprise about 0.1% of body mass.

③ Critical for biological functions:-

(i) growth.

(ii) Oxygen transport

(iii) metabolism

(iv) Disease Defence.

④ Imbalances, either excess or deficiency, over time can result in severe illness or death.

The End.

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Chemical Reactions!

Ch₂O₄

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Chemical reactions:

⇒ A process that occurs when two or more molecules interact to make a new product



Types of chemical reaction:

1. Precipitation reaction (କ୍ଷତ୍ପତ୍ତି-କ୍ରିୟା)

2. Acid-base reaction (ଅମ୍ଳ-ବେସ-କ୍ରିୟା)

3. Oxidation-reduction reaction (ଓକ୍ସିଡେସନ୍-ରିଡକ୍ସନ୍-କ୍ରିୟା)

Precipitation Reaction (କ୍ଷତ୍ପତ୍ତି-କ୍ରିୟା)

⇒ A precipitation reaction is one in which dissolved substance react to form one or more solid product.
(କ୍ଷତ୍ପତ୍ତି-କ୍ରିୟା ହେଉଛି ଏକ ପ୍ରକାର ରାସାୟନିକ ପ୍ରକ୍ରିୟା ଯେଉଁଠି ଦ୍ରବଣୀୟ ପଦାର୍ଥମାନଙ୍କର ମିଶ୍ରଣ ଯୋଗୁଁ ଏକ କଠିନ ପଦାର୍ଥ (କ୍ଷତ୍ପତ୍ତି) ଗଠିତ ହୁଏ।)



In this reaction AgCl is Insoluble.

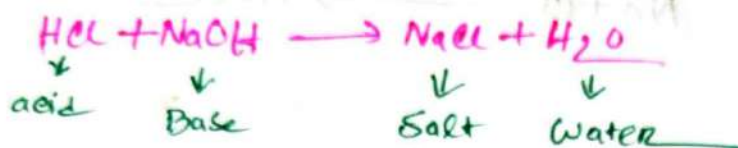
TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Acid-Base Reactions!

⇒ It is a chemical reaction that occurs between an acid and a base. While the removal of a hydrogen ion from the acid and its addition to the base.



Note:- Acid react with Base to form salt and water.

Example of Acid & Base

Name of Acid

Present in

1. Acetic acid (CH_3COOH) → Vinegar

2. Citric acid ($\text{C}_6\text{H}_8\text{O}_7$) → Lemon juice.

3. Hydrochloric acid (HCl) → Digestive fluid of the stomach

Name of Base

Used in/Prezent in

1. Aqueous Ammonia (NH_4OH)/ NH_3 → Household cleaner.

2. $\text{CH}_3\text{-OH}$ → Car window washer

3. Ca(OH)_2 → Slaked lime (For Building construction)

4. Mg(OH)_2 → Milk of magnesia.

5. NaOH → Drain/oven cleaners.

Simple property of Acid and Base

1. Acid - Base Indicators:-

- (i) Used to distinguish between acidic and basic solutions through color changes.
- (ii) Natural materials like tea and red cabbage juice act as Indicators.
- (iii) Litmus turns red in acidic and blue in basic solution.
- (iv) phenolphthalein is colorless in acidic and pink in basic solution.
- (v) Lemon juice lightens tea (amber color) due to citric acid.
- (vi) Red cabbage juice changes from red to green & yellow with Base addition, and reverses with acid addition.

TOPIC NAME : _____

Definition of Acids & Bases:—

⇒ There are Three Concepts, acid and base.

1. Arrhenius Concept

2. Bronsted-Lowry, Concept

3. Lewis Concept.

Arrhenius Concepts:—

"Acid" is a compound that gives H^+ ions in H_2O .

"Base" is a compound that gives OH^- ions in H_2O .



→ Here, HCl is an acid, since it release/gives H^+ ions in water when dissolved in H_2O .

→ Here, $NaOH$ is a base, since it gives OH^- ions in the water when dissolved in H_2O .

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Usefulness:-

⇒ This concept is useful in the study of chemical reactions.

Limitations:-

1. In non-aqueous solutions, the theory fails to explain the acidic or basic nature of the substance.

2. Some base do not contains OH^- ions also some acid do not contains H^+ ions then it fails to explain the acidic or basic nature of the substance.

Ex:- $\text{SO}_2, \text{CO}_2, \text{SO}_3, \text{NH}_3, \text{CaO}$.

Bronsted-Lowry Concept

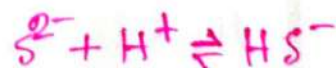
"Acid" - is a substance which donates an H^+ ions or proton.

"Base" - is a substance which accepts an H^+ ions or proton.

Acid:-



Base:-



TOPIC NAME : _____

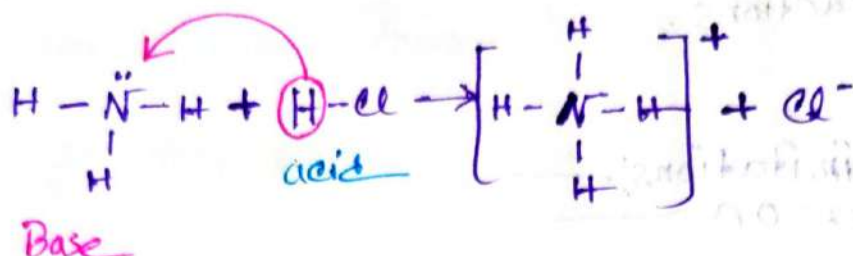
DAY : _____

TIME : _____

DATE : / /

Reactions:

①



Here, in this reaction NH_3 is a Base, Since NH_3 accepts a H^+ ions during the reaction.

on the other hand HCl is an Acid, Since HCl donates a H^+ ions during the reaction.

②



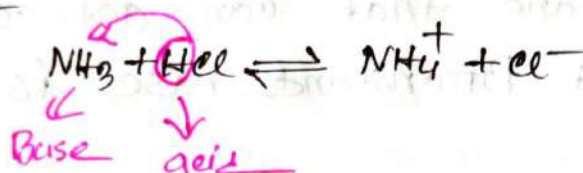
Here CaO is a Base, Because of CaO accept H^+ ions and H_2O is an Acid Because of H_2O donates H^+ ions during the reaction process.

Q:- Why Bronsted-Lowry Concept is Superior to Arrhenius Concept?

⇒ (1) It allows us, for a wider range of substances to be classified as acids or bases.

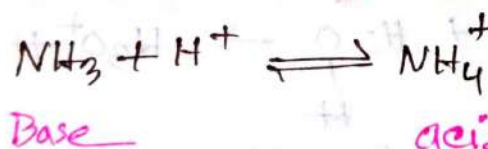
(2) Not limited to aqueous solutions.

Ex:-



(3) Release of OH^- ions not necessary to qualify as a Base

Ex:-



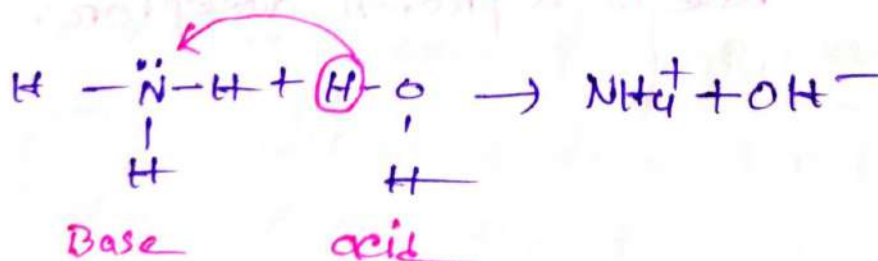
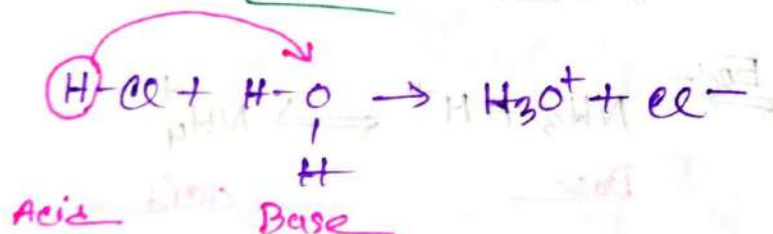
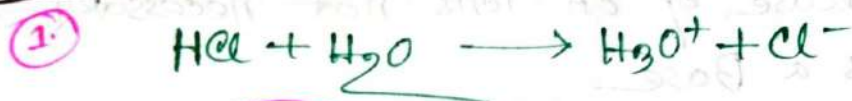
Note:- An acid is a proton donor, while a base is a proton acceptor.

Amphiprotic Substance:-~~अम्लीय और क्षारीय~~

⇒ A substance that can act as both acids and base by being both proton acceptor and proton donor.

OR

⇒ A substance that can act as both, Brønsted acid and Brønsted base is called Amphiprotic Substance.

Ex:-

Here, in the 1st reaction H_2O accepts a H^+ ions that's why according to Brønsted concept H_2O is Base.

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

and in the 2nd reaction H_2O donate a H^+ ions that's why According to Brønsted Concept H_2O is a Acid.

So, since H_2O can acts as both, Brønsted acid and Brønsted Base, then we can called H_2O is a Amphiprotic substance.

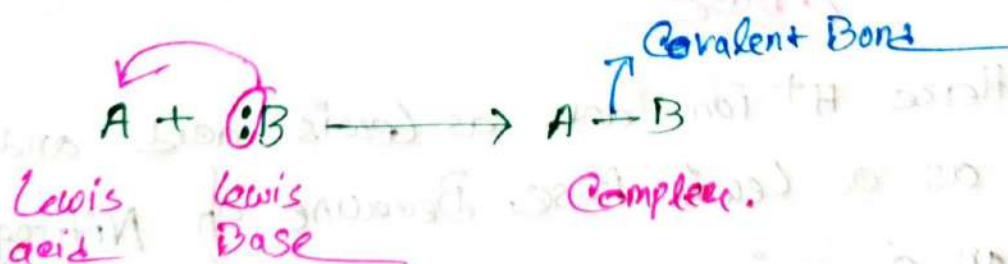
Other examples of As:-

1. HS^- , $Al(H_2O)_3(OH)_3$ etc.

Lewis Concept:-

"Acid" is a substance which can Accept a pair of e^- .

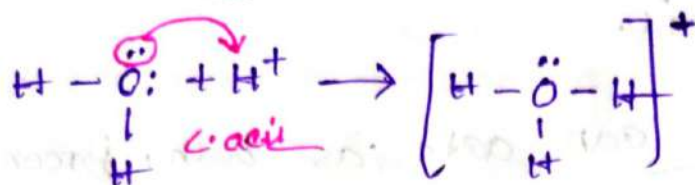
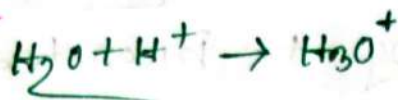
"Base" is a substance which can Donates a pair of e^-



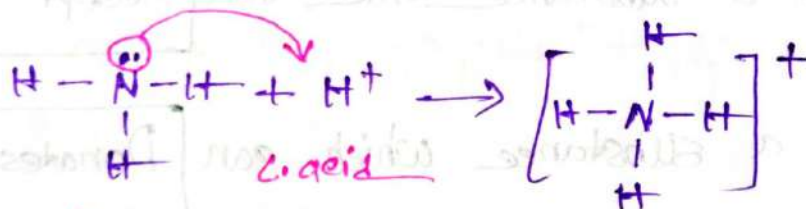
TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Example: 1:-L. BaseL. acid

Here H^+ ion acts as a Lewis acid and H_2O acts as a Lewis Base. Because of Oxygen atom donates an e^- pair to the proton (H^+). Forming a Coordinate Covalent bond in the process.

Example: 2:-L. acidL. Base

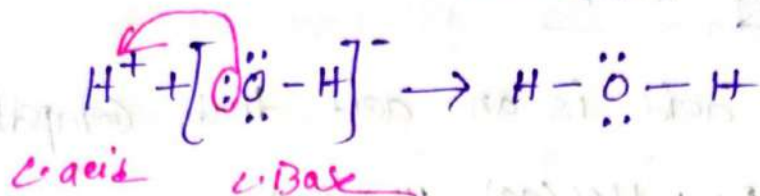
Here H^+ ion acts as Lewis acid and NH_3 acts as a Lewis Base. Because of Nitrogen atom donates an e^- pair to the proton (H^+). Forming a Coordinate Covalent bond during the process.

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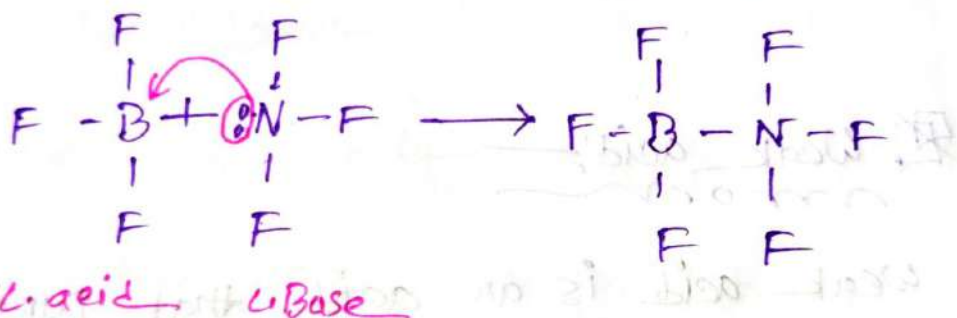
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Example 3:



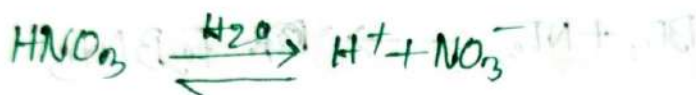
Example 4:



Strong and Weak Acids and Bases :-

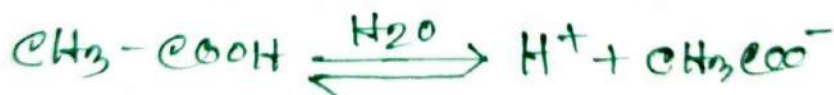
Strong acid :-

⇒ A Strong acid is an acid that completely ionize in Water. Ex: $\text{HCl}(\text{aq})$, $\text{HNO}_3(\text{aq})$



Weak acid :-

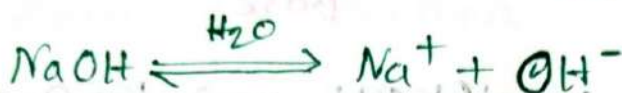
⇒ A weak acid is an acid that partially ionizes in Water. Ex: $\text{CH}_3\text{-COOH}$, $\text{HCN}(\text{aq})$, $\text{HF}(\text{aq})$



Strong Base:-

⇒ A base that is completely ionized in water.

Ex:- NaOH , Ca(OH)_2 , KOH



Weak Base:-

⇒ A base that is only partly ionized in water.

Ex:- NH_3 , Mg(OH)_2 , Na_2CO_3 , NaHCO_3 .



Neutralization Reactions:-

- acid + base

⇒ The reactions of an acid and a base, which forms water and a salt, is called Neutralization Reaction.

OR

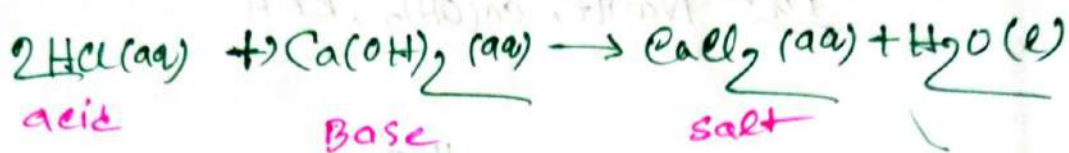
⇒ A reaction of an acid and a base that results in an ionic compound called salt and possibly water, is called Neutralization reaction.

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

⇒ (যে স্বাভাবিক বিক্রিয়ায় একটি অম্ল ও একটি ক্ষার বিক্রিয়া করে নিরপেক্ষ লবন এবং পানি তৈরি করে তাকে ন্যূনতম বিক্রিয়া বলে)



Here the 1st reaction, the Base: Ca(OH)_2 supplies Ca^{2+} cations, the acid HCl supplies Cl^- anions. The salt contains Ca^{++} and Cl^- ions (CaCl_2).

In the 2nd reactions, the acid H_2SO_4 supplies SO_4^{2-} anions, the Base KOH supplies K^+ cations. The salt contains SO_4^{2-} and K^+ ions (K_2SO_4).

Oxidation-Reduction Reaction:-

(ऑक्सीकरण-प्रचरण-प्रतिक्रिया)

⇒ It commonly known as redox reactions, are reactions that involve the transfer of electrons from one species to another.

❖ Oxidation (ऑक्सीकरण)

⇒ Oxidation is the Loss of electrons
(ऑक्सीकरण में e^- का ह्रास होता है)

❖ Reduction (प्रचरण)

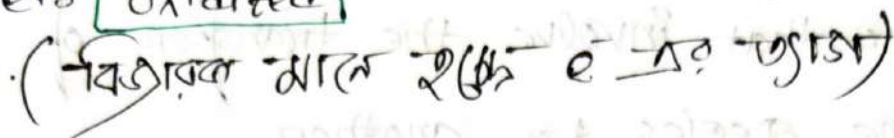
⇒ Reduction is the gains of electrons
(प्रचरण में e^- का अर्जन होता है)

❖ Oxidation Agent (ऑक्सीकरण प्रचरण) / Oxidant

⇒ An Oxidation Agent or Oxidant, gains electrons and reduced in a chemical reaction. It gets reduced.
(ऑक्सीकरण प्रचरण में e^- का अर्जन होता है)

Reducing Agent (Rakshak)

⇒ An Reducing Agent, losing electrons and it gets oxidized.



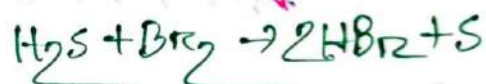
Note:-

1. Oxidation meant (i) Addition of oxygen. $\rightarrow \text{C} + \text{O}_2 \rightarrow \text{CO}_2$

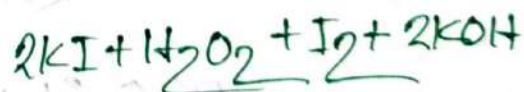
(ii) Addition of electronegative element



(iii) Removal of Hydrogen



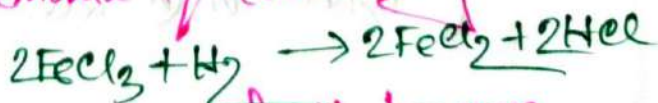
(iv) Removal of electropositive element



2. Reduction meant (i) Removal of oxygen



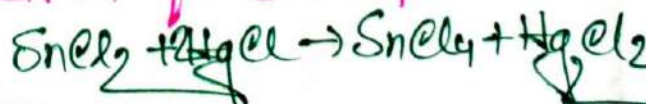
(ii) Removal of electronegative element



(iii) Addition of Hydrogen



(iv) Addition of electropositive element

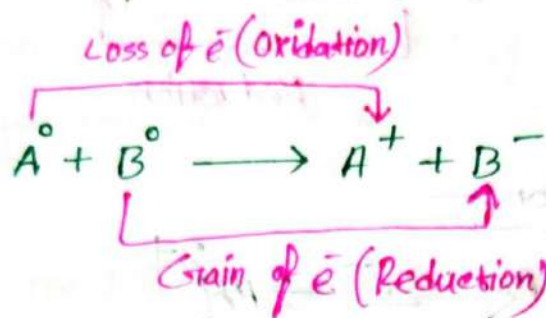


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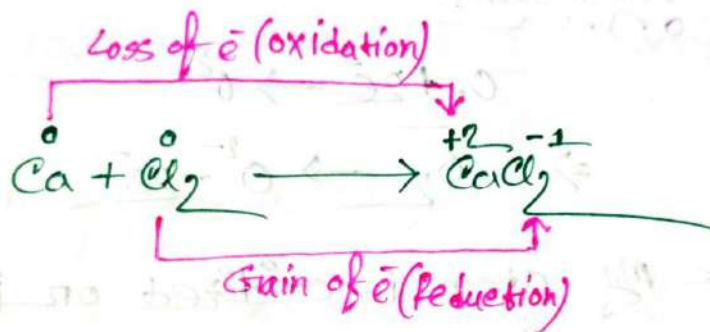
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Oxidation and Reduction In terms of Electron Transfer



example! :-
mon



Here, Ca atom is oxidized since it increase in oxidation Number or it loss $2e^-$. And Cl is reduced since it decreases in oxidation number (0 to -1) or it gains $1e^-$.

Fact:-
mon

⇒ Oxidation and Reduction reaction are always interlinked. Because e^- are neither created or destroyed in a chemical reaction. It occurs in pairs, It is impossible to have one without the other.

⇒ Oxidation is the half-reaction in which there is a loss of e^- .

⇒ Reduction is another half-reaction in which there is a gain of e^- .

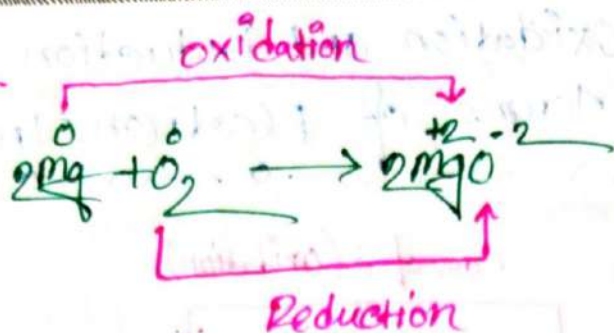
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DAY : _____

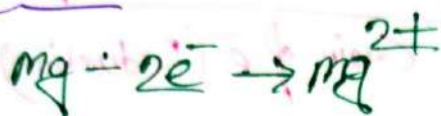
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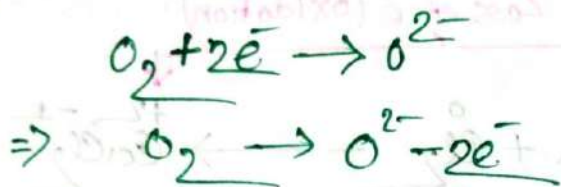
Example: 2:-



Half-oxidation:-



Half-Reduction:-



$\Rightarrow$ Here Mg atom is oxidized or Reducing Agent since it increases its oxidation Number (0 to 2) or loss 2e^- .
 And O_2 is called Reduced or Oxidation Agent since it decreases its Oxidation Number (0 to -2) or gains 2e^- .

TOPIC NAME : _____

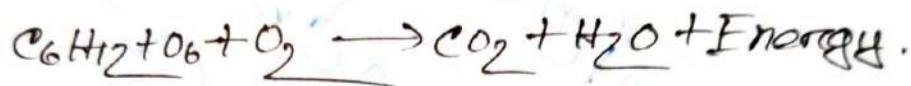
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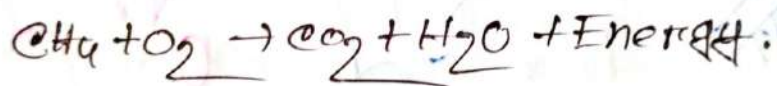
Important of Oxidation-Reduction:-

1. Photosynthesis involves reducing CO_2 to produce $(\text{C}_6\text{H}_{12}\text{O}_6)$ Carbohydrates + O_2 + H_2O + Sunlight

2. Cellular respiration oxidizes nutrients in cell to generate energy.



3. Combustion of Fossil Fuels



4. Reduction of metal oxides, like iron ores,



TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

#Chemical Bonds

Di-o

Chemical Bonds :-

⇒ It is defined as a strong attractive force that exists between two or more atoms to hold them together as a stable molecule in a substance. (স্থায়ীভাবে বন্ধিত একটি পদার্থ, অথবা বা অণুর মধ্যে একটি স্থায়ী আকর্ষণ যা স্থায়ীভাবে বন্ধিত করে।)

Definitions to Understand Chemical Bonding :-Valence Electron (বাহ্যিক ইলেকট্রন) :-

⇒ The electrons in the outer most energy level in an atom are called Valence Electron or Valency electron. (কোন কোনো নির্দিষ্ট প্রকারের পরমাণুর বাহ্যিক সর্বোচ্চ শক্তিস্তরে অবস্থিত ইলেকট্রনকে বাহ্যিক ইলেকট্রন (Valency) ইলেকট্রন বা বাহ্যিক ইলেকট্রন বলে।)

Note:- "Last outer most energy level are also called Valency shell"

(-এই সর্বোচ্চ শক্তিস্তরকে বাহ্যিক ইলেকট্রন বলে।)

Bonding electrons :-

⇒ The valence electrons actually involved in bond are called Bonding electrons.

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Octet rule: (ଅକ୍ଟ ନିୟମ):

⇒ In chemical bond formation, atoms interact by losing, by gaining or by sharing electrons to acquire a stable Noble gas Configuration. The tendency of atoms to prefer to have eight(8) electrons in the valance shell, are known as Octet rule.

(ବିଭିନ୍ନ ଲୋକେ ପଦାର୍ଥ ସମ୍ବନ୍ଧରେ ନିଜେ ଯେ ନଷ୍ଟ ହେବାକୁ ଚାହୁଁଛନ୍ତି ଏବଂ ଅନ୍ୟତମ ସହ ଯୋଗାଯୋଗ କରିବାକୁ ଚାହୁଁଛନ୍ତି ତାହା ହେଉଛି ଅକ୍ଟ ନିୟମ। ଏହା ପଦାର୍ଥର ସ୍ଥିରତା ବୃଦ୍ଧି କରିବା ପାଇଁ ଏକ ନିୟମ।)

Valence and Valency

Valence: (ମୂଲ୍ୟ):

⇒ The no. of electrons in the outermost shell of a metallic element and the odd no. of electrons in the outer-most shell of a non-metallic element are called valance.

(ଯଦି ଏକ ଲୋକେ ନିଜେ ନିଜେ କଣ୍ଠପଥରେ ଏ ସଂଖ୍ୟା ଗୁଣ କରନ୍ତି ଏବଂ ଏହାକୁ ମୂଲ୍ୟ କୁହାଯାଏ। ଏହା ପଦାର୍ଥର ସ୍ଥିରତା ବୃଦ୍ଧି କରିବା ପାଇଁ ଏକ ନିୟମ।)

Valency (ব্যালেন্সি) :-

⇒ The total No. of electrons in the outermost shell of an element is called valency of that element.

(একটি মৌলিক পদার্থের বাহ্যিক কক্ষের মোট ইলেকট্রন সংখ্যাকে বোঝায়)

Types of Bond :-

Main types of Bond :-

1. Ionic or electrovalent bond, (আয়নিক বন্ধন)

2. Covalent Bond (সমন্বিত বন্ধন)

3. Coordinate Covalent Bond (সম্মিলিত বন্ধন)

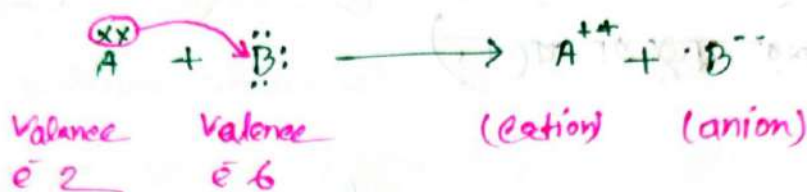
Others Important Bond :-

1. Metallic Bond (ধাতব বন্ধন)

2. Hydrogen Bond (হাইড্রোজেন বন্ধন)

Ionic Bond:-

⇒ Ionic Bond is the complete transfer of valance $e(s)$ between atoms.



Cation:-

⇒ The atom that loses $e(s)$ are called cation (positive ion)

Anion:-

⇒ The atom that gains $e(s)$ are called anion (Negative ion)

Conditions for Formation of Ionic Bond:-

- ① Bond Formed between metals and non-metals.
- ② Metal is Donor (cation), Non-metal is Acceptor (Anion)
- ③ Bond are not Formed between two metals or two non-metals.

③ Net Lowering Energy.

Eg:- To form a stable Ionic compound, there must be a net lowering of the energy.

Let, calculate the Energy of formation of NaCl molecule:-



So Net energy = $187 + 85 - 119$
= 153 kcal .

⑤ Electro-negativity difference between the atoms:-

$$\text{Na} \rightarrow 0.9$$

$$\text{Cl} \rightarrow 3.0$$

$$\therefore \text{Diff} = 3 - 0.9 = 2.1$$

So, it's possible

because of minimum difference needed 1.7

Electro-negativity of 2nd and 3rd periods:-

2nd Periods:-

Li	Be	B	C	N	O	F	Ne
1.0	1.5	2.0	2.5	3.0	3.5	4.0	0

3rd Periods:-

Na	Mg	Al	Si	P	S	Cl	Ar
0.9	1.2	1.5	1.8	2.1	2.5	3.0	0

Factors that Controlled the Formation of Ionic Bond:-

1. Ionization energy (आयनन शक्ति)
2. Electron Affinity (इलेक्ट्रॉन आकर्षण)
3. Lattice Energy

Example of Ionic Bond:-

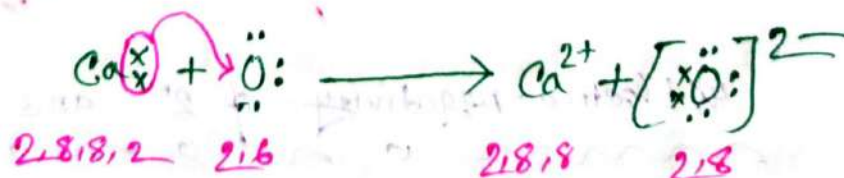
Example 1:-

1. NaCl



Example 2:-

2. CaO



Characteristics of Ionic Compounds:-

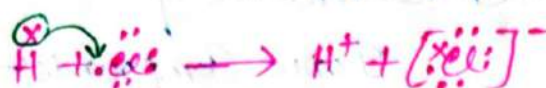
- 1 Solids at room temperature.
- 2 High melting and boiling point.
- 3 Hard and brittle.
- 4 Soluble in water but Insoluble in organic solvents.
- 5 Conductors of electricity.
- 7 Don't Exhibit Isomerism.
- 8 Undergo ionic reaction which are fast.

Q:- Explain Dot and cross diagram?

⇒ This Diagrams can be used to illustrate the Bonds. A dot and cross diagram can show electrons being shared between two non-metals where electrons from one atom are shown as dot and electrons from another atom as crosses.

OR:-
⇒ Dot and cross Diagrams are diagrams that show the Arrangement of the Outer-shell $e(s)$ in an Ionic or covalent Compound.

Ex:-

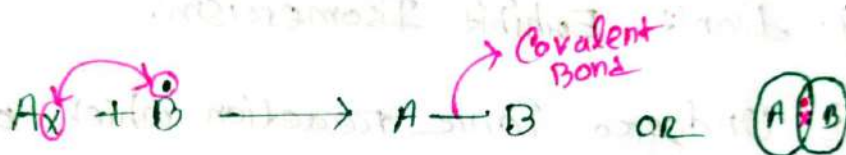


'Dot' $\rightarrow$ represent the outer e^- of one type of atom (Cl).

'Cross' $\rightarrow$ represent the outer e^- of the other type of atom (H)

Covalent Bond :-

$\Rightarrow$ Covalent Bond is the sharing of electrons between two or more atoms.



Condition for Formation of Covalent Bond :-

1. Bond formed between two, Same or different Non-metals
2. Equal Electronegativity.
3. Equal sharing of e^- (s)

TOPIC NAME : _____

DAY : _____

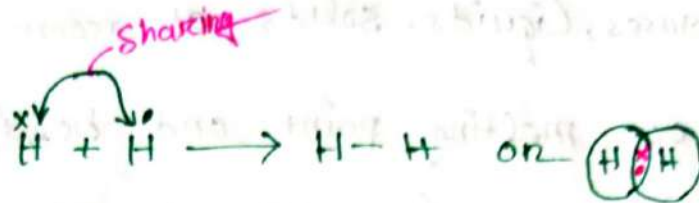
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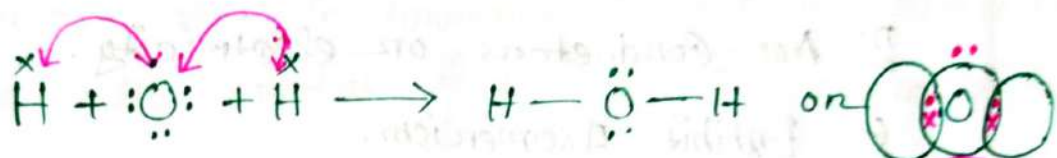
Example of Covalent Compounds:-

Example:-

1. H_2



2. H_2O



Unshared $e^-(s)$

3. O_2



4. CO_2



Characteristics of

Covalent compounds:-

1. Gases, Liquids, solids at room temp.
2. Low melting point and boiling point
3. Soft, much readily broken.
4. Insoluble in water but soluble in organic soln.
5. Non-conductors of electricity.
6. Exhibit Isomerism.

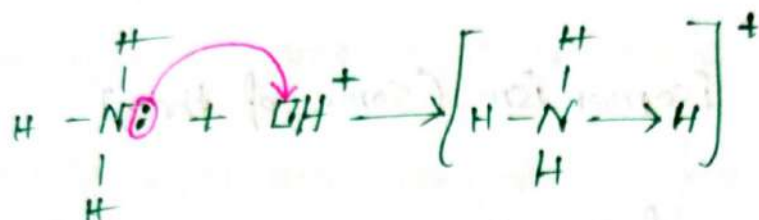
Q:- Why you think something can conduct electricity

⇒ The ability of a material to conduct electricity is linked to the presence of free charge carriers (e^- or ions) and their mobility within the material.

Coordinate Covalent Bond :-

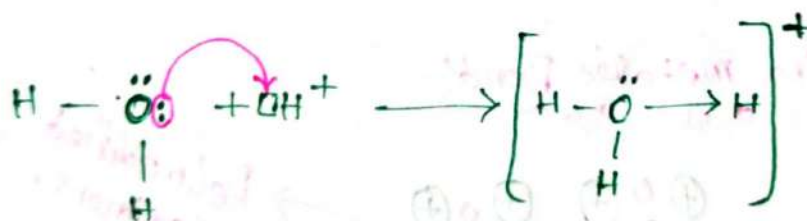
⇒ Coordinate covalent Bond is formed by sharing of an electron pair from a single atom.

1. NH_4^+ :-



Here, the Nitrogen atom in Ammonia donates its electron pair to the empty orbital of the H^+ ion. Since Nitrogen is the donor, H^+ is the acceptor, and coordinate bond is formed.

2. H_3O^+ :-



Here, the Oxygen atom in water donates its electron pair to the empty orbital of the H^+ ion. Since Oxygen is the donor, H^+ is the acceptor and coordinate bond formed.

Properties of Coordinate Compounds:-

1. They have lower melting or boiling points than ionic compound.
2. Exhibit isomerism (some of them)
3. Sharing of e^- (s) takes place in definite direction.
4. It is weaker than ionic bonding.

Metallic Bonding:-

⇒ Metallic Bond is a bond that holds atoms together in a metallic substance.

Ex of Na metallic Bond:-

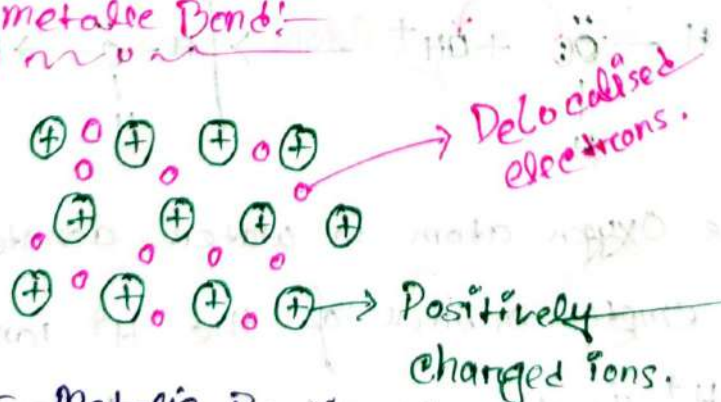
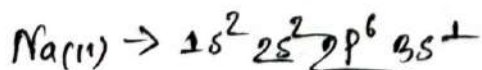


Fig:- Metallic Bonding in Sodium



⇒ Hence Na contains one e^- in its valance shell. In the solid state, metallic sodium features an array of Na^+ ions

TOPIC NAME : _____

DAY : _____

TIME : _____

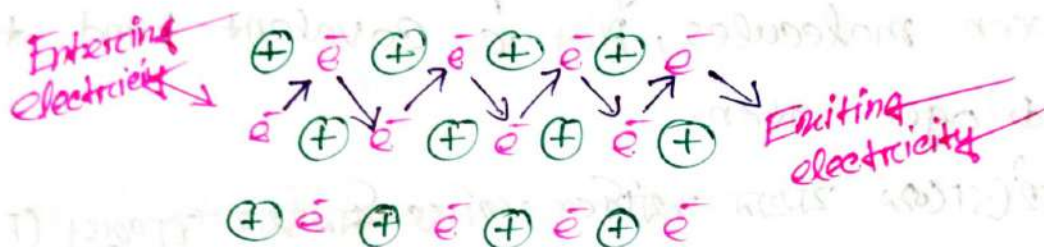
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that are surrounded by a sea of s orbital.

Electrical Conductivity:

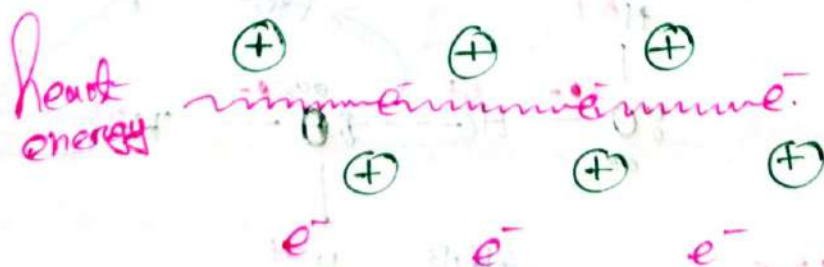
$\Rightarrow$ It can be defined as how much voltage is required that to get an amount of electric current to flow.

The mobile $e^{-}(s)$ are responsible for electric conductivity.



Heat Conductivity:

$\Rightarrow$ It can be defined as the ability of a material to conduct heat from its one side to the other.



Hydrogen Bond:-

⇒ An hydrogen bond is an attraction Between two atoms that already participate in other chemical Bonds.

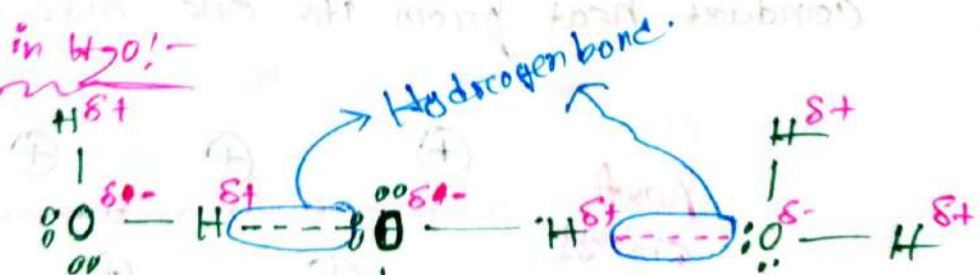
OR

⇒ It is a special type of dipole-dipole attraction between molecules, Not a covalent bond, to a Hydrogen atom.

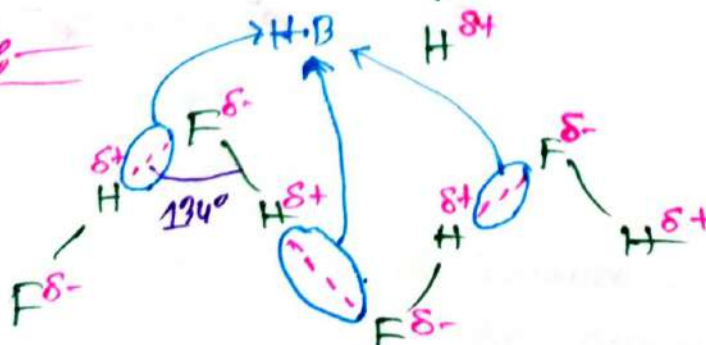
(-শর্তপ্রত্যয় যখন অধিক তড়িৎ ঋণাত্মক প্রাণী (F, O, N) এর সাথে যুক্ত হয় তখন প্রান্তের তড়িৎ ঋণী হয়, যা প্রান্তের তড়িৎ ঋণী সমূহ H এর সাথে অধিক তড়িৎ ঋণাত্মক প্রাণীর যে কোন তড়িৎ ঋণী তাকে বলা হয় শর্তপ্রত্যয় বিন্দু,

Ex:-

H-Bond in H₂O:-



In HF:-



Bond Length:

⇒ The Distance between the the centers of two covalently bonded atom

Note:-

1. If No. of bond Increases, bond length decreases due to more attraction.

Bond Angle:

⇒ The Angle Between two Bonds.

it depends:-

- ① charge distribution
- ② geometry of the molecules.
- ③ Symmetry.
- ④ Hybridization.

Bond Energies:

⇒ Its defined as the energy required to split the molecule into atoms.

it depends:-

on the type of bonding.

- ① Single
- ② Double.
- ③ Triple.

#Molecular Orbital Theory

Ch: 06

#Molecular Orbital Theory (MOT):-

⇒ In Simple terms, The MOT states that each atom tends to combine together and form molecular Orbitals.

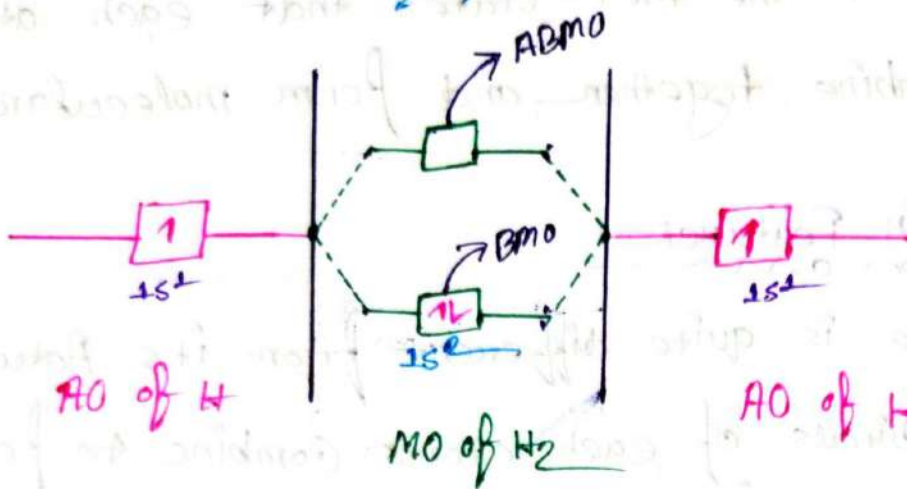
Main Features:-

1. A molecule is quite different from its Actual Atom.
2. Atomic orbitals of each atoms combine to form molecular orbital.
3. Molecular Orbital also follows the all rules of Atomic Orbital.
4. The molecular Orbital have different energy level.
5. The shapes of molecular orbital formed depend upon the shape of combining atomic orbital.

Some Shortcut:-

1. Atomic Orbital's = AO's
2. Molecular Orbital's = MO's
3. Bond Order = B.O
4. Bonding Molecular Orbital = B.M.O
5. Anti Bonding Molecular Orbital = ABMO

Formation of H_2 molecule:-



Bond Order (BO):-

⇒ The Number of bonding pairs of electrons between two atoms.

Mathematically we can write:-

$$B.O = \frac{BMO - ABMO}{2} = \frac{N_b - N_a}{2}$$

Notes:-

- ① if $N_b > N_a$ The molecule is stable that's means bond possible.
- ② if $N_b \leq N_a$ The molecule is unstable that's means no bond possible.

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

#Q. Draw the Hydrogen Molecule Orbital Diagram and Also Find

1. Bond Order

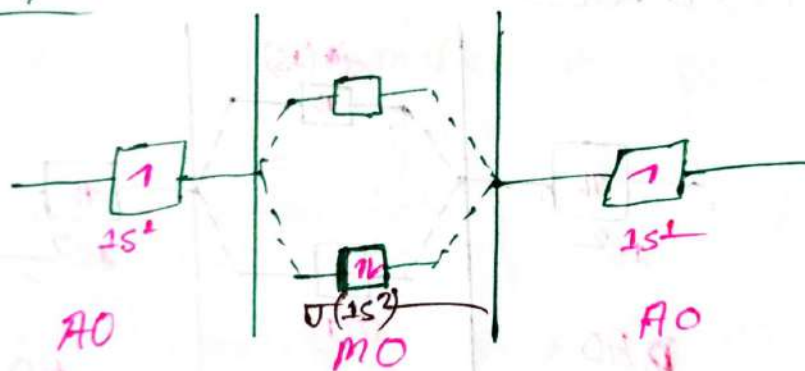
2. Magnetic properties.

3. Electronic Configuration:-

⇒ We know:-

The H atom e^- configuration:-
 $1s^2$

So, the MO:-



$$\textcircled{1} \quad B.O = \frac{N_b - N_a}{2} = \frac{2 - 0}{2} = 1$$

② Magnetic properties : Diamagnetic, Since No unpaired e^- in BMO

③ e^- configuration : $KK \sigma(1s)^2$

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

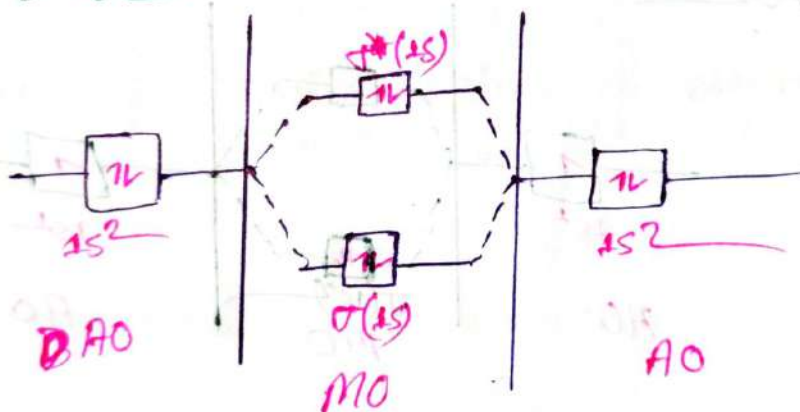
#Q: Draw the He MO Diagram
and Also find

- (i) B.O
- (ii) Magnetic properties (if any have)
- (iii) e^- configuration (if have any)

⇒ we know:-

The He atom e^- configuration:-
 $1s^2$

So, the M.O Diagram:-



(1) $B.O = \frac{N_b - N_a}{2} = \frac{2 - 2}{0} = 0$; therefore He molecule is not formed

(2) None Since $B.O = 0$

(3) e^- configuration = $KK \sigma(1s)^2 \sigma^*(1s)^2$ (if He₂ formed)

Notes:-

Bond dissociation energy.

⇒ Greater the Bond Order (B.O) greater is the bond dissociation energy.

Bond Length (BL):-

⇒ B.O is inversely proportional to the BL

$$\therefore BL \propto \frac{1}{BO}$$

→ Higher the BO, smaller the BL.

Magnetic properties:-

① If unpaired electron present in MO's then we call it paramagnetic

Ex:- O_2 , NO, Cr, etc.

② If there is no unpaired e^- in MO's it will be called diamagnetic

Ex:- N_2 , CO, etc.

Molecular Orbitals Diagram

Homo-nuclear Molecules

1. Q:- Draw the N_2 Molecular Orbital Diagram and Find.

(i) Bond Order.

(ii) Magnetic properties

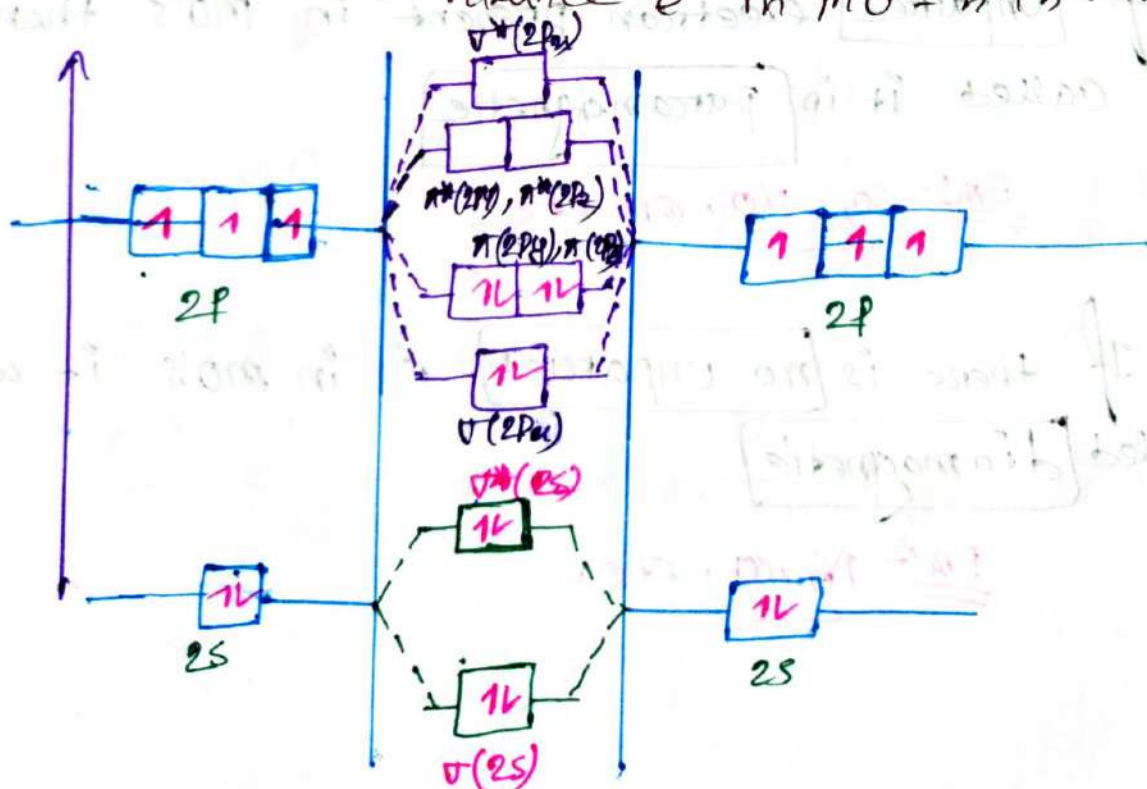
(iii) Electronic Configuration.

⇒ We know -

e^- configuration of $N(7) = 1s^2 2s^2 2p_x^1 2p_y^1 2p_z^1$

Valance e^- for one atom of $N = 5$

∴ So total Valance e^- in MO = $5 + 5 = 10$



TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

⇒ we know

(i)

$$\text{Bond Order (B.O)} = \frac{N_b - N_a}{2}$$

$$= \frac{8 - 2}{2}$$

= 3 ; three covalent exist.

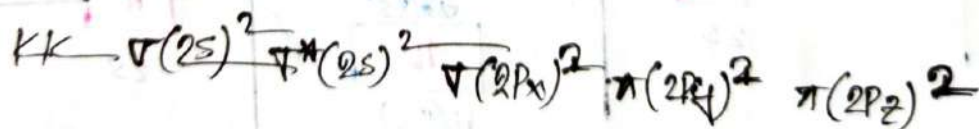
(ii)

⇒ Magnetic properties : Diamagnetic,

Since there is no unpaired e^- in MO

(iii)

⇒ Electronic Configuration:-



TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Q:2 Draw the O_2 molecular and Find

Orbital Diagram.

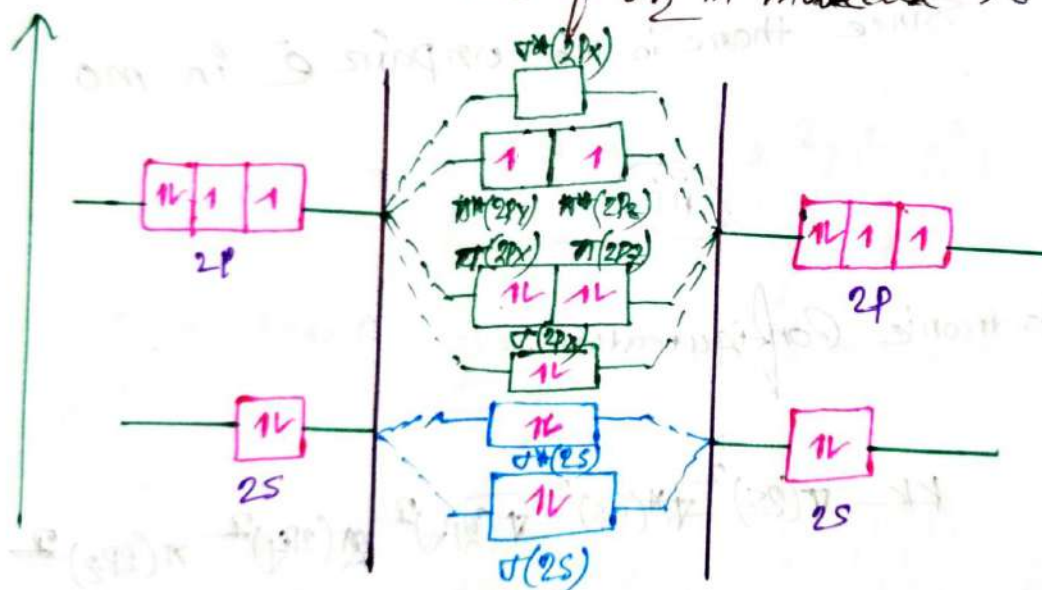
- (i) B.O
- (ii) Magnetic properties
- (iii) e^- configurations.

$\Rightarrow$ We know:-

e^- configuration of O_2 (8) $\rightarrow 1s^2 2s^2 2p_x^2 2p_y^1 2p_z^1$

The Valance e^- of one oxygen atom $\rightarrow 6$

$\therefore$ total Valance e^- of O_2 in molecule $\rightarrow 6+6=12$



$\Rightarrow$

We know:-

$$BO = \frac{M_b - M_a}{2} = \frac{8 - 4}{2} = 2 ; \text{two covalent bond exist}$$

Magnetic properties:- Paramagnetic; Since there is 2 unpaired e^- in MO

e^- configuration: $KK \sigma(2s)^2 \sigma^*(2s)^2 \sigma(2p)^2 \pi(2p_x)^2 \pi(2p_y)^2 \pi^*(2p_x)^1 \pi^*(2p_y)^1$

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Q:3:- Draw the molecular Orbital and Find

(a) Bond order

(b) Magnetic properties.

(c) σ configuration of Neon molecule (Ne_2)

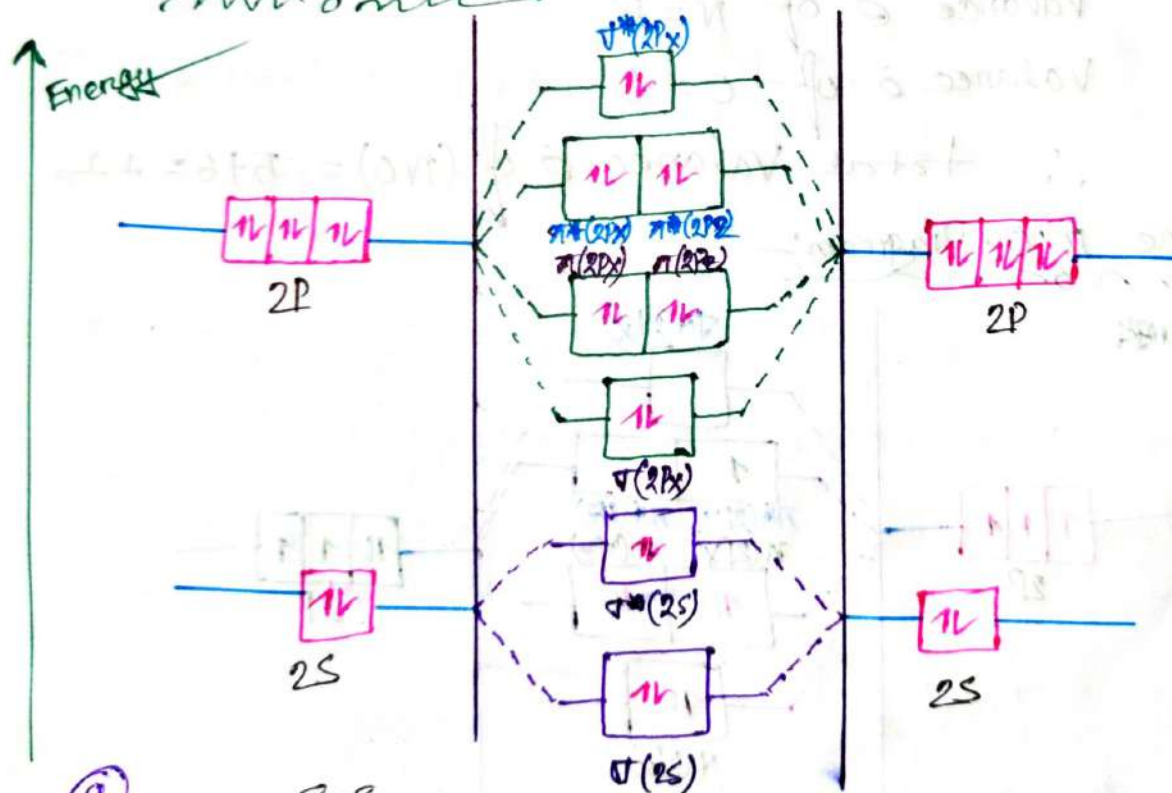
$\Rightarrow$ We know:-

$\rightarrow$ The e^- configuration of $Ne(10) \rightarrow 1s^2 2s^2 2p_x^2 2p_y^2 2p_z^2$

So, Balance $e^- = 8$

$\therefore$ total Balance e^- of Ne_2 molecule $= 8 + 8 = 16$

Now:- the MO Diagram:-



(a) $B.O = \frac{8-8}{2} = 0$, therefore Ne_2 molecule not formed.

(b) have No magnetic properties Since it does not form molecule.

For Heteronuclear Molecules:-

Q:1— Draw the Nitric-oxide (NO) molecular Orbital Diagram and Find.

- ① Bond Order
- ② Magnetic properties.
- ③ e^- Configuration:

$\Rightarrow$ We know:-

e^- configuration of $N(7) = 1s^2 2s^2 2p_x^1 2p_y^1 2p_z^1$

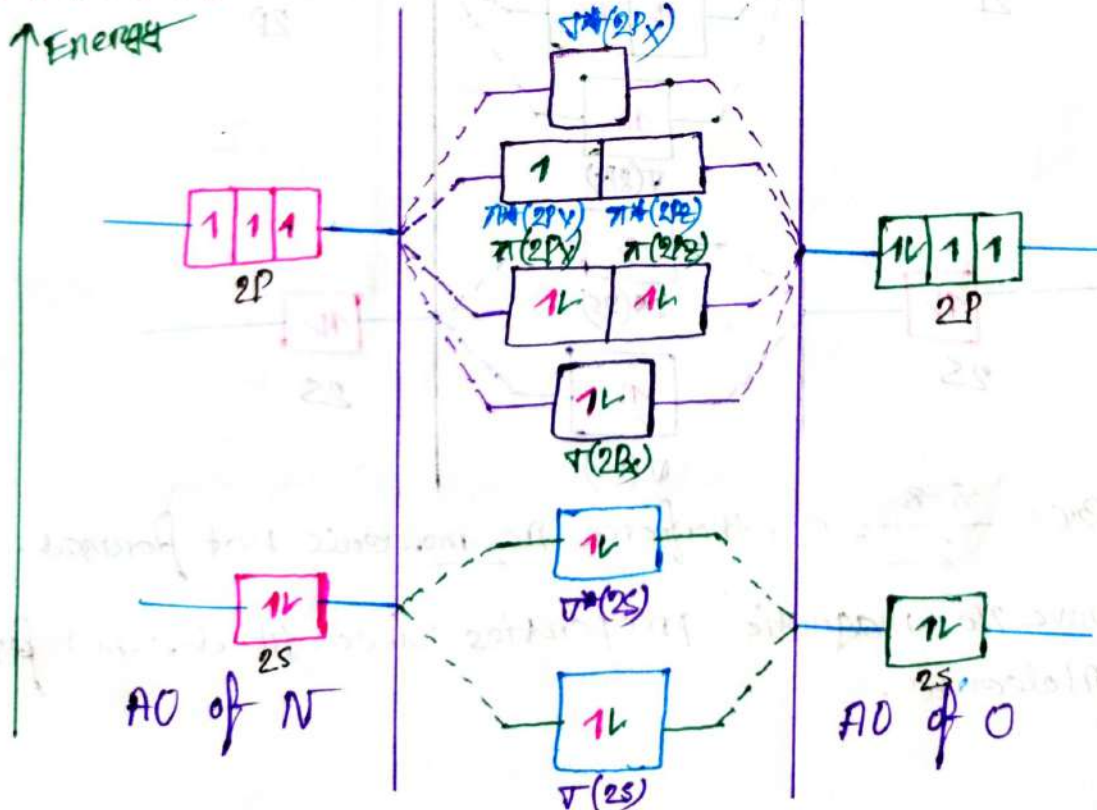
$$O(\delta) = 15^4 25^2 28^2 29^4 22^1$$

Valance e^- of $N=5$.

Balance of $\frac{1}{2} \cdot 0 = 0$

$\therefore$ total variance of $(\pi_0) = 5+6 = 11$

So, the M.O. Diagram:-



(mod of no)

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

(a) $B.O = \frac{8-3}{2} = 2.5$

(b) magnetic properties :- **Paramagnetic** ; Since there is one unpaired e^- in MO's

(c) e^- configuration :-



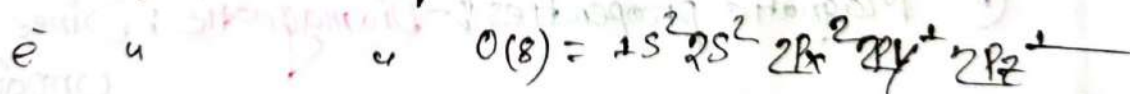
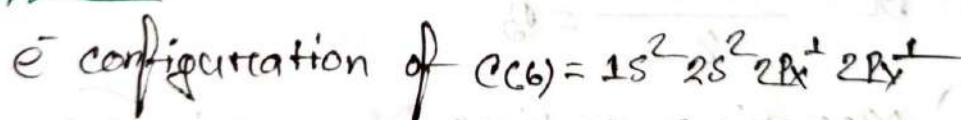
Q:2:- Draw the Carbon-monoxide (CO) molecular Orbital Diagram and Find

(1) Bond Order.

(2) Magnetic properties.

(3) e^- configurations :-

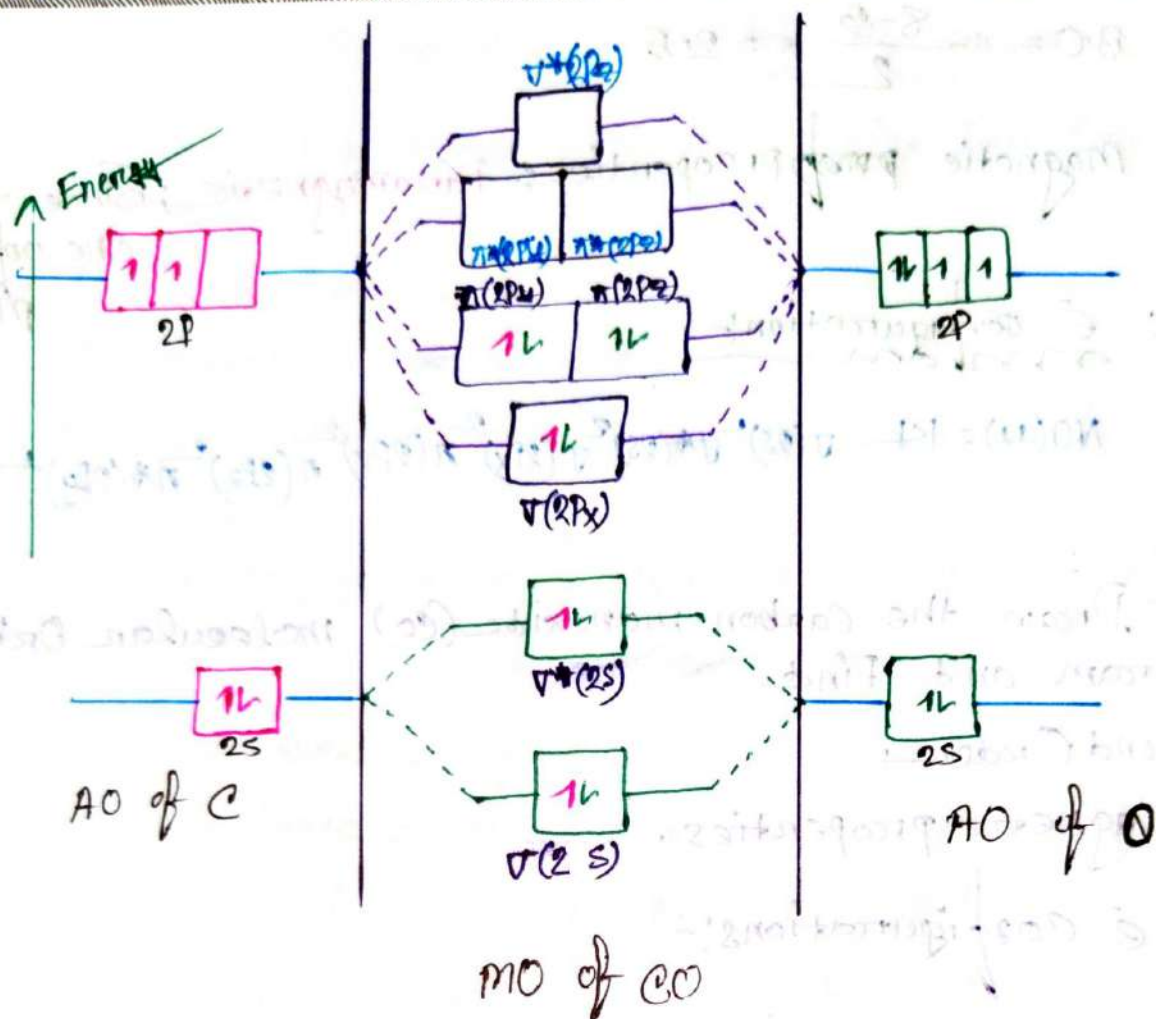
$\Rightarrow$ We know :-



Valence e^- of $C = 4$

" " e^- of $O = 6$

total No of Valance e^- for (CO) = 10



a) $BO = \frac{8-2}{2} = 3$

b) Magnetic properties :- **Diamagnetic**; Since there is no unpaired e^- in MO's

c) e^- configuration:—

